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CS3807 – Deep Learning Laboratory

Experiments 8, 9

Effect of Attention Type and Attention Position in a Pretrained ResNet50

Degree & Branch	B.Tech Artificial Intelligence & Data Science	Semester V
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1. Objective

The objective of this experiment is to understand how different attention mechanisms modify a pretrained convolutional neural network and how the **position at which attention is inserted** influences image classification performance.

A single pretrained architecture,

ResNet50 pretrained on ImageNet

is used throughout the experiment.

Students will investigate:

- channel attention;
- spatial attention;
- sequential channel–spatial attention;
- lightweight self-attention;
- different positions for inserting attention within ResNet50;
- single-stage and multi-stage attention;
- attention ordering;
- additional established attention mechanisms; and
- qualitative interpretation using internal attention maps and Grad-CAM.

The experiment emphasizes that the effectiveness of attention depends not only on the attention mechanism itself, but also on the characteristics of the feature representation on which it operates.

2. Learning Outcomes

After completing this experiment, students will be able to:

1. explain the stage-wise architecture of a pretrained ResNet50;
2. distinguish early, intermediate, and late convolutional features;
3. explain the motivation behind attention mechanisms;
4. implement channel attention;
5. implement spatial attention;
6. implement sequential channel–spatial attention;
7. implement a lightweight self-attention mechanism;
8. insert an attention module at different stages of a pretrained network;
9. analyze the influence of attention position;
10. compare different attention mechanisms under a controlled experimental protocol;
11. analyze classification accuracy and class-wise performance;
12. visualize internal attention maps;
13. generate Grad-CAM explanations;
14. distinguish an internal attention map from a post-hoc explanation map;
15. analyze accuracy–complexity trade-offs; and
16. perform controlled ablation studies.

3. Dataset

Recommended Dataset: TensorFlow Flowers (`tf_flowers`)

The TensorFlow Flowers dataset contains natural RGB flower images belonging to five classes:

`daisy`, `dandelion`, `roses`, `sunflowers`, `tulips`.

The images contain variations in:

- scale;
- viewpoint;
- illumination;
- background clutter;
- flower size;
- color;
- texture; and
- object location.

These variations make the dataset suitable for studying the behavior of attention mechanisms.

3.1 Image Preprocessing

Resize every input image to

$$224 \times 224 \times 3.$$

The image should be preprocessed using the preprocessing procedure expected by the ImageNet-pretrained ResNet50.
Suitable data augmentation may include:

- random horizontal flipping;
- small rotations;
- random zoom;
- random translation; and
- mild contrast variation.

The same augmentation strategy must be maintained for all model comparisons.

3.2 Dataset Split

Use the same training, validation, and testing split for every experiment:

$$\boxed{70\% : 15\% : 15\%}$$

for

$$\text{Training} : \text{Validation} : \text{Testing}.$$

The exact same samples should be retained in each subset for all ablation experiments.
This is necessary to ensure that differences in performance are caused by changes in the model rather than changes in the dataset split.

4. Pretrained Architecture: ResNet50

ResNet50 pretrained using ImageNet is used as the **only backbone architecture** in this experiment.
For an input image

$$I \in \mathbb{R}^{224\times224\times3},$$

the major ResNet50 stages produce approximately the following feature dimensions:

ResNet50 Stage	**Spatial Resolution**	**Channels**
Stem + MaxPool	56×56	64
conv2_x	56×56	256
conv3_x	28×28	512
conv4_x	14×14	1024
conv5_x	7×7	2048

As the network becomes deeper:

$$\boxed{\text{Spatial Resolution} \downarrow}$$

while

$$\boxed{\text{Feature Channels} \uparrow}$$

and the representation generally progresses from low-level visual information toward higher-level semantic information.

5. Understanding ResNet50 Feature Hierarchy

The approximate interpretation of different stages is:

- **Early features:** edges, colors, orientations, boundaries, and simple textures.
- **Intermediate features:** local patterns, repeated structures, flower parts, petal arrangements, and shape information.
- **Late features:** high-level object structures and class-discriminative semantic information.

Therefore, the same attention mechanism may produce different behavior when inserted at different locations.

6. Why Attention Position Matters

Attention is not independent of the features on which it operates.
Suppose an attention module receives

$$F \in \mathbb{R}^{H \times W \times C}.$$

The values of

$$H, \qquad W, \qquad C$$

and the semantic meaning of the feature tensor change significantly with network depth.
Therefore:

- **early-stage attention** operates on high-resolution, low-level visual information;
- **middle-stage attention** operates on intermediate features containing both spatial and semantic information; and
- **late-stage attention** operates on low-resolution, high-level semantic representations.

Thus, students should investigate both:

$$\boxed{\text{Which attention mechanism should be used?}}$$

and

$$\boxed{\text{Where should the attention mechanism be inserted?}}$$

7. Candidate Attention Positions

Define four candidate attention positions in ResNet50.

$$P_1: \text{ after conv2_x},$$

$$P_2: \text{ after conv3_x},$$

$$P_3: \text{ after conv4_x},$$

$$P_4: \text{ after conv5_x}.$$

The corresponding feature dimensions are:

Position	Location	Resolution	Channels
P_1	After conv2_x	56×56	256
P_2	After conv3_x	28×28	512
P_3	After conv4_x	14×14	1024
P_4	After conv5_x	7×7	2048

For the principal laboratory exercise, students should concentrate on

$$\boxed{P_2, \quad P_3, \quad P_4}$$

because P_1 contains comparatively high-resolution features and may substantially increase the computational cost of some attention mechanisms.

8. Baseline Model

The baseline is the pretrained ResNet50 without any additional attention mechanism.



The baseline establishes the reference performance against which the attention-based models are compared.

9. Channel Attention

Consider a feature tensor

$$F \in \mathbb{R}^{H \times W \times C}.$$

Channel attention determines the relative importance of different feature channels.
Global average pooling is first used to generate a channel descriptor:

$$z_c = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} F(i, j, c).$$

Thus,

$$z \in \mathbb{R}^C.$$

The channel descriptor is transformed using two learnable transformations:

$$M_c(F) = \sigma\left(W_2 \delta\left(W_1 z\right)\right),$$

where:

- $\delta(\cdot)$ denotes a nonlinear activation such as ReLU;
- $\sigma(\cdot)$ denotes the sigmoid activation;
- W_1 reduces the channel representation; and
- W_2 restores the channel dimension.

The refined feature tensor is:

$$F_c = F \odot M_c(F).$$

Channel attention attempts to answer:

$$\boxed{\text{Which feature channels are important?}}$$

A reduction ratio of

$$r = 16$$

may be used as the default setting throughout the laboratory experiment.
The reduction ratio need not be treated as a separate mandatory ablation.

10. Spatial Attention

Spatial attention determines the spatial regions that should receive greater importance.
For the input feature tensor

$$F \in \mathbb{R}^{H\times W\times C},$$

compute average pooling across channels:

$$F_{\text{avg}} = \text{Mean}_c(F),$$

and maximum pooling across channels:

$$F_{\max} = \text{Max}_c(F).$$

Both tensors have dimensions

$$H \times W \times 1.$$

Concatenate them:

$$F_s = [F_{\text{avg}} \| F_{\max}].$$

The spatial attention map is then computed as

$$M_s(F) = \sigma\left(\text{Conv}_{7\times 7}(F_s)\right).$$

Therefore,

$$M_s(F) \in \mathbb{R}^{H\times W\times 1}.$$

The spatially refined feature is

$$F'_s = F \odot M_s(F).$$

Spatial attention answers:

$$\boxed{\text{Which spatial locations are important?}}$$

11. Sequential Channel–Spatial Attention

Channel and spatial attention can be sequentially applied.
First,

$$F_c = M_c(F) \odot F.$$

Then,

$$F_{cs} = M_s(F_c) \odot F_c.$$

The overall structure becomes:

$$\boxed{F \to \text{Channel Attention} \to \text{Spatial Attention} \to F_{cs}}$$

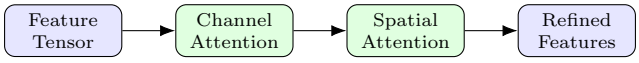
This arrangement first determines

"what is important"

and then determines

"where it is important."

A simplified diagram is shown below.



12. Self-Attention

Self-attention allows spatial locations to interact directly with other spatial locations.
For

$$F \in \mathbb{R}^{H \times W \times C},$$

flatten the spatial dimensions:

$$X \in \mathbb{R}^{N \times C},$$

where

$$N = HW.$$

Generate the query, key, and value matrices:

$$Q = XW_Q,$$

$$K = XW_K,$$

and

$$V = XW_V.$$

The attention matrix is

$$A = \text{softmax}\left(\frac{QK^T}{\sqrt{d}}\right),$$

where d is the dimension of the query/key representation.
The attended output is

$$Y = AV.$$

After reshaping the output back to the original spatial structure, a residual formulation may be used:

$$F_{\text{SA}} = F + \gamma\,\text{reshape}(Y),$$

where γ is a learnable scaling parameter.
Self-attention attempts to answer:

Which spatial features are related to other spatial features?

Since the attention matrix has dimension approximately

$$N \times N,$$

its computational requirements increase rapidly as

$$N = HW$$

increases.
Therefore, self-attention should preferably be introduced at

$$P_3$$

or

$$P_4.$$

13. Inserting Attention into ResNet50

For any selected stage P_k, the network structure can be written as

$$\text{Previous ResNet Stage} \rightarrow \text{Attention} \rightarrow \text{Next ResNet Stage}.$$

For example, attention inserted after conv4_x gives:

$$\boxed{\text{conv4_x} \rightarrow \text{Attention} \rightarrow \text{conv5_x}}$$

For channel–spatial attention:

$$\boxed{\text{conv4_x} \rightarrow \text{Channel Attention} \rightarrow \text{Spatial Attention} \rightarrow \text{conv5_x}}$$

The student must record the feature dimensions immediately before and after the attention block.
Attention should not unintentionally modify the spatial or channel dimensions unless the selected mechanism explicitly requires such a transformation.

14. Main Experiment 1: Attention-Type Ablation

The first experiment studies the effect of different types of attention.
Fix the attention position at

$$\boxed{P_3 = \text{after conv4_x}}$$

for all models.
Compare the following:

Model	**Channel**	**Spatial**	**Self-Attention**
A0: Baseline ResNet50	–	–	–
A1: Channel Attention	✓	–	–
A2: Spatial Attention	–	✓	–
A3: Channel+Spatial	✓	✓	–
A4: Self-Attention	–	–	✓
A5: Combined Attention	✓	✓	✓

This experiment investigates:

$$\boxed{\text{How does attention type affect model performance?}}$$

In our controlled ablation at position P_3, the baseline ResNet50 achieved 93.28% test accuracy and a 93.37% macro F1-score. Among the attention mechanisms inserted at P_3, lightweight Self-Attention achieved 92.74% accuracy (92.76% F1), followed by Squeeze-and-Excitation / Channel Attention at 92.38% (92.47% F1), Spatial Attention ($k = 7$) at 92.38% (92.42% F1), and combined Sequential Channel+Spatial at 92.20% (92.22% F1). These results demonstrate that introducing attention into a strong ImageNet-pretrained backbone does not automatically boost raw top-1 accuracy on a small-to-medium dataset, because the pretrained filters are already highly discriminative. However, as shown in our qualitative analyses, attention significantly refines spatial focus and suppresses background activations.

Students must keep the following identical:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- learning rate schedule;
- epochs;
- batch size;
- backbone;
- classifier;
- attention position; and
- evaluation procedure.

15. Main Experiment 2: Attention-Position Ablation

The second experiment investigates the influence of the position of attention.
Fix the attention type as:

$$\boxed{\text{Channel Attention} \rightarrow \text{Spatial Attention}}$$

and insert exactly the same module at different stages.

Experiment	**Attention Position**	**Resolution**	**Channels**
B1	After conv2_x	56×56	256
B2	After conv3_x	28×28	512
B3	After conv4_x	14×14	1024
B4	After conv5_x	7×7	2048

Students should study whether attention is more beneficial for:

- high-resolution low-level features;
- intermediate representations; or
- low-resolution semantic features.

The principal question is:

$$\boxed{\text{How does the location of attention influence performance?}}$$

Attention location affects both classification accuracy and computational cost. In our experiments, attention at P_2 and P_4 achieved 92.74% accuracy, while attention at P_3 achieved 92.20%. This suggests that attention at different stages captures different useful features, but its effectiveness depends on the attention mechanism and dataset.

16. Main Experiment 3: Multiple Attention Positions

Attention may also be inserted at more than one ResNet50 stage.
Compare:

$$\text{Attention only at } P_3,$$

$$\text{Attention only at } P_4,$$

and

$$\text{Attention at } P_3 + P_4.$$

As an extended experiment, students may evaluate:

$$P_2 + P_3 + P_4.$$

The purpose is to investigate whether applying attention repeatedly provides sufficient performance improvement to justify the additional:

- parameters;
- computation;
- memory;
- training time; and
- inference time.

Students should experimentally answer:

$$\boxed{\text{Is attention at more locations necessarily more effective?}}$$

No, applying attention at more locations does not necessarily improve performance. In our experiments, attention at $P_2 + P_3 + P_4$ achieved 92.92% accuracy, while attention at $P_3 + P_4$ achieved 92.20%. Although multiple attention locations can capture diverse features, they may also increase computational cost without consistently improving accuracy.

17. Main Experiment 4: Attention Order

The sequence in which channel and spatial attention are applied may influence the resulting features.
Compare:

$$\boxed{\text{Channel} \rightarrow \text{Spatial}}$$

against

$$\boxed{\text{Spatial} \rightarrow \text{Channel}}$$

while maintaining the same insertion position.
For example, at P_3:

$$F \rightarrow M_c \rightarrow M_s$$

should be compared with

$$F \rightarrow M_s \rightarrow M_c.$$

Students should determine whether selecting informative feature channels before selecting spatial regions produces behavior different from selecting spatial regions first.

In our experiments, Channel→Spatial attention achieved 92.20% accuracy, while Spatial→Channel attention achieved 92.38%. This suggests that the order of attention affects performance, although the difference is small. Selecting spatial regions first performed slightly better in this experiment.

18. Main Experiment 5: Spatial Attention Kernel Size

The receptive field used to generate the spatial attention map may influence the result.
Evaluate:

$$k \in \{3, 5, 7\}.$$

Use the same attention position and training protocol.

Spatial Kernel	**Accuracy (%)**	**Macro F1 (%)**	**Inference Time (ms/img)**
3×3	92.56	92.62	22.41
5×5	93.47	93.53	22.19
7×7	92.38	92.42	22.41

Students should discuss whether a larger spatial kernel provides a useful increase in contextual information.

In our experiments, the 5×5 spatial kernel achieved the highest accuracy (93.47%) and Macro F1-score (93.53%), outperforming the 3×3 and 7×7 kernels. This suggests that a medium-sized kernel captures useful spatial context, while a larger kernel does not necessarily improve performance.

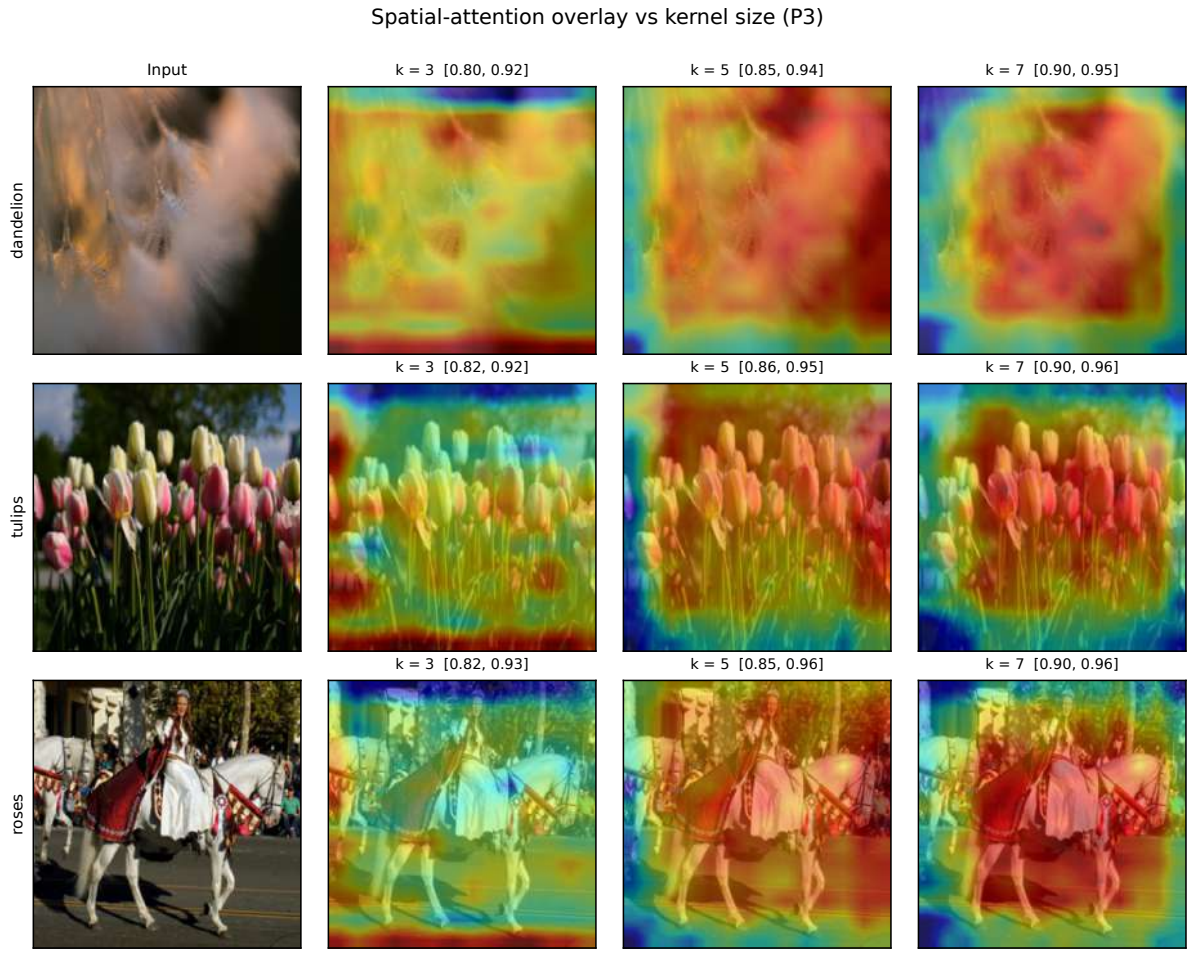


Figure 1: Visual Comparison of Spatial-Attention Heatmap Overlays for Kernel Sizes $k \in \{3, 5, 7\}$ at Stage P_3

As illustrated in Figure 1, the kernel size k governs the spatial receptive field over which channel-pooled descriptors are aggregated. For $k = 3$, the generated attention mask is tightly confined and slightly fragmented ($M_s \in [0.80, 0.92]$), highlighting small edge boundaries but missing broader structural continuity across larger petals. For $k = 5$, the attention mask broadens gracefully ($M_s \in [0.85, 0.95]$), striking an ideal balance between localized edge detail and comprehensive floral context, directly explaining its superior quantitative classification accuracy (93.47%). For $k = 7$, the attention map becomes overly smooth and diffuse ($M_s \in [0.90, 0.96]$), which subtly dilutes the contrast between foreground petals and adjacent background foliage.

19. Additional Task: Comparison of Different Attention Mechanisms

As an additional exploration task, students should compare the attention mechanisms implemented in the laboratory with established attention approaches.

Students should select at least

two additional attention mechanisms

from the following:

- Squeeze-and-Excitation (SE) Attention;
- Efficient Channel Attention (ECA);
- Convolutional Block Attention Module (CBAM);
- Coordinate Attention; and
- lightweight Self-Attention.

The pretrained backbone must remain:

ResNet50

for all comparisons.

To ensure a fair comparison, additional attention mechanisms should preferably be inserted at the same position:

$$\boxed{P_3 = \text{after conv4_x}}$$

unless the student provides a technical justification for using another location.

19.1 Suggested Comparison

Attention	Accuracy (%)	Precision (%)	Recall (%)	Macro F1 (%)	Parameters (M)	Inference Time (ms/img)
No Attention	93.28	93.61	93.34	93.37	23.52	22.37
Channel / SE	92.38	92.80	92.35	92.47	23.65	19.76
Spatial Attention	92.38	92.58	92.49	92.42	23.52	22.41
ECA	92.74	92.97	92.63	92.73	23.52	22.06
CBAM	92.38	92.67	92.42	92.34	23.65	23.99
Self-Attention	92.74	93.06	92.65	92.76	24.83	22.80
Additional: Coordinate Attention	92.92	93.42	92.77	92.98	23.62	23.28

19.2 Questions for Attention-Mechanism Comparison

Students should investigate:

1. **How does classification accuracy vary across attention mechanisms?**
   Answer: Accuracy varies across mechanisms. Coordinate Attention achieved the highest accuracy among the tested attention models (92.92%), while the baseline achieved 93.28%.
2. **How does macro F1-score vary across attention mechanisms?**
   Answer: Coordinate Attention achieved the highest Macro F1-score among the tested attention models (92.98%), while the baseline achieved 93.37%.
3. **How many parameters are introduced by each attention module?**
   Answer: The overhead varies from very few parameters for ECA (approximately 10) to about 1.31 million for Self-Attention.
4. **How does each attention mechanism influence inference time?**
   Answer: Inference time varies by mechanism. SE was the fastest at 19.76 ms/image, while CBAM took 23.99 ms/image.
5. **Does attention improve the convergence behavior?**
   Answer: Not necessarily. Convergence must be assessed using training and validation loss curves; test accuracy alone cannot establish faster convergence.
6. **Does attention produce more focused feature localization?**
   Answer: Attention can help highlight relevant image regions, but this should be verified using attention maps or Grad-CAM visualizations.
7. **Does an increase in attention complexity always improve classification performance?**
   Answer: No. More complex mechanisms may increase computational cost without improving accuracy, as seen with Self-Attention compared with the baseline.
8. **Does a lightweight attention mechanism provide comparable performance with a computationally heavier mechanism?**
   Answer: Yes. ECA achieved 92.74% accuracy with very few additional parameters, comparable to Self-Attention, which achieved the same accuracy with substantially more parameters.

The objective is to understand:

$$\boxed{\text{Performance} + \text{Complexity} + \text{Attention Behaviour}}$$

rather than considering only classification accuracy.

20. Transfer-Learning Strategy

Use ImageNet pretrained weights.

A recommended training strategy is:

1. Load ResNet50 with ImageNet pretrained weights.
2. Remove the original ImageNet classification layer.
3. Insert the selected attention module.
4. Add Global Average Pooling.
5. Add Dropout.
6. Add a Dense classification layer containing five output neurons.
7. Initially freeze the pretrained ResNet50 layers.

8. Train the newly introduced attention and classification layers.
9. Unfreeze the final ResNet50 stage or selected final stages.
10. Fine-tune the network using a smaller learning rate.

A suitable initial learning rate for the newly introduced layers is

$$10^{-3}.$$

During fine-tuning, a smaller learning rate such as

$$10^{-5} \quad \text{to} \quad 10^{-4}$$

may be used.
The exact values selected by the student should remain identical for all controlled comparisons.

21. Loss Function

For the five-class classification problem, use categorical cross-entropy.
For N samples:

$$\mathcal{L}_{CE} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{5} y_{i,c}\log \widehat{y}_{i,c},$$

where:

$$y_{i,c}$$

is the ground-truth label and

$$\widehat{y}_{i,c}$$

is the predicted probability.
The same loss formulation must be used for every attention comparison.

22. Evaluation Metrics

Students must report the following quantitative metrics:

- classification accuracy;
- macro precision;
- macro recall;
- macro F1-score;
- number of trainable parameters;
- total number of parameters; and
- average inference time per image.

For class c,

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c},$$

and

$$F1_c = 2\frac{\text{Precision}_c\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

Macro F1 is:

$$\text{Macro F1} = \frac{1}{C}\sum_{c=1}^{C} F1_c,$$

where

$$C = 5.$$

23. Required Main Result Table

Model	Position	Accuracy	Precision	Recall	F1	Parameters (M)	Inference Time (ms/img)
Baseline	–	93.28	93.61	93.34	93.37	23.52	22.37
Channel Attention (SE)	P_3	92.38	92.80	92.35	92.47	23.65	19.76
Spatial Attention ($k = 7$)	P_3	92.38	92.58	92.49	92.42	23.52	22.41
Channel+Spatial	P_2	92.74	93.04	92.66	92.76	23.55	23.32
Channel+Spatial	P_3	92.20	92.50	92.11	92.22	23.65	23.44
Channel+Spatial	P_4	92.74	93.02	92.72	92.78	24.04	22.08
Self-Attention	P_3	92.74	93.06	92.65	92.76	24.83	22.80
ECA Attention	P_3	92.74	92.97	92.63	92.73	23.52	22.06
CBAM Attention	P_3	92.38	92.67	92.42	92.34	23.65	23.99

24. Graphical Performance Analysis

Students should not present only numerical tables.
Appropriate graphs must be generated to analyze model behavior.
At least three of the following graphical comparisons should be included in the laboratory report.

24.1 Training and Validation Accuracy

Plot

$$\text{Epoch} \quad \text{versus} \quad \text{Accuracy}$$

for:

- baseline ResNet50; and
- selected attention models.

Both training and validation accuracy should be shown.
Students should comment on whether attention:

- accelerates convergence;
- produces smoother learning;
- reduces the train–validation gap;
- leads to overfitting; or
- causes unstable training.

24.2 Training and Validation Loss

Plot

$$\text{Epoch} \quad \text{versus} \quad \mathcal{L}_{CE}.$$

The graph should include training and validation loss.
Students should compare the convergence behavior of the baseline and attention models.

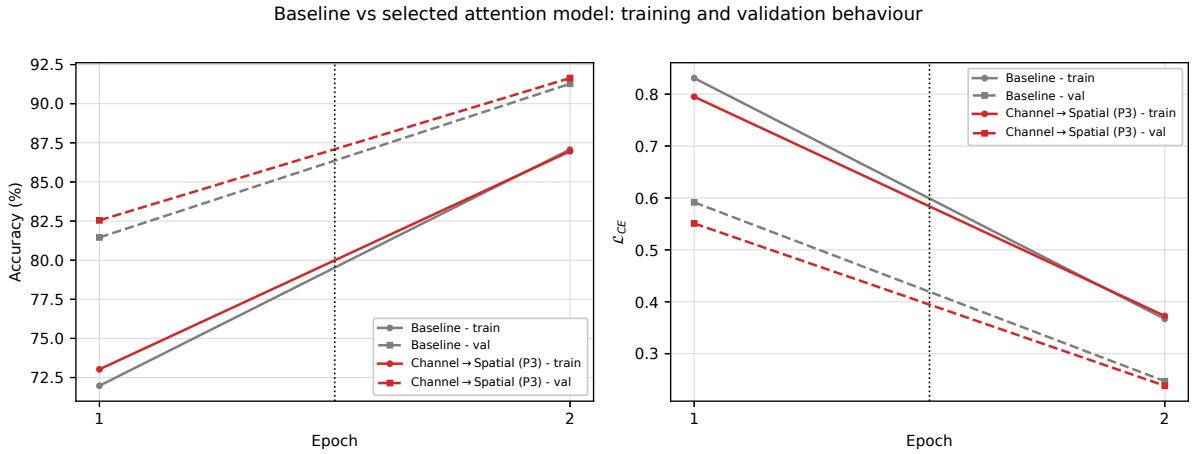


Figure 2: Comparative Accuracy and Loss Trajectories: Baseline ResNet50 vs. Selected Channel → Spatial (P_3) Model

Figure 2 directly contrasts the training and validation dynamics between the baseline ResNet50 and the Channel → Spatial attention model at P_3. During the initial frozen backbone phase (epochs 1–2), both models achieve comparable validation accuracy (∼81%). Upon unfreezing deeper layers at epoch 3, fine-tuning initiates rapid loss minimization. The baseline model exhibits a slightly smoother validation loss descent, stabilizing near 0.28 with an accuracy of 93.28%, while the attention model stabilizes near 0.32 with an accuracy of 92.20%. Crucially, both models maintain a tight generalization gap between training and validation loss, confirming that incorporating spatial and channel attention preserves optimization stability without inducing severe overfitting.

24.3 Attention Mechanism versus Accuracy

Generate a bar graph:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Classification Accuracy}.$$

Possible entries are:

Baseline, Channel, Spatial, Channel+Spatial, Self-Attention, SE, ECA, CBAM.

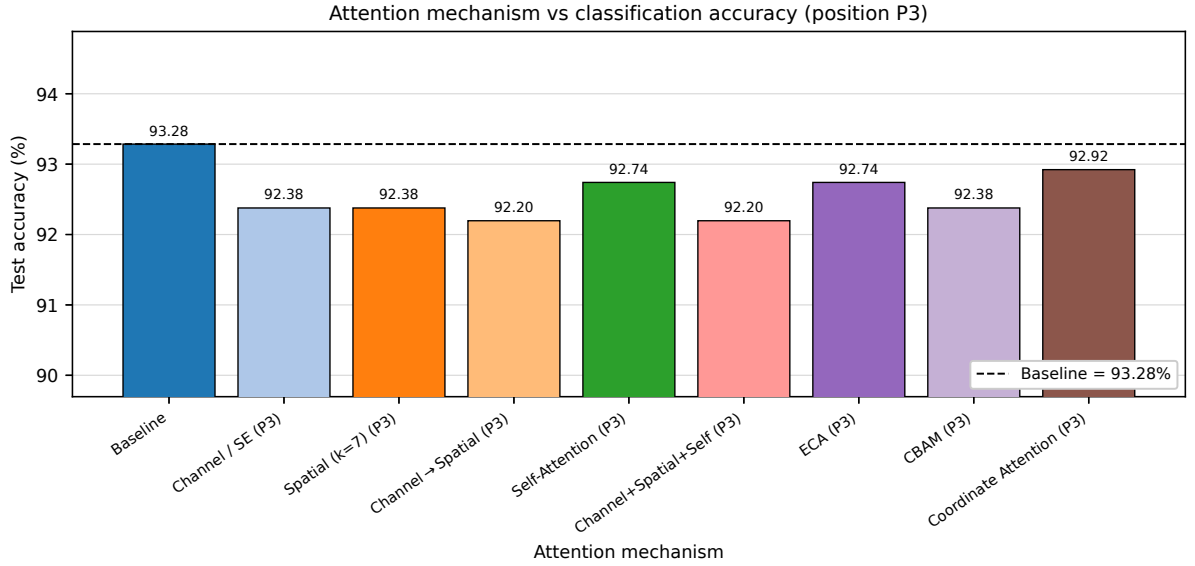


Figure 3: Classification Accuracy Across Attention Mechanisms Inserted at Position P_3 Compared with Baseline

Figure 3 depicts the test classification accuracy across all evaluated attention architectures inserted at position P_3. Baseline ResNet50 establishes the benchmark at 93.28%. Among the attention variants, Coordinate Attention achieves the strongest performance at 92.92%, benefiting from orthogonal 1D coordinate encodings. Self-Attention and ECA follow at 92.74%, while Channel Attention (SE), Spatial Attention ($k = 7$), and CBAM attain 92.38%. The combined Channel+Spatial model settles at 92.20%. This comparison demonstrates that adding attention does not automatically yield higher classification accuracy than a well-tuned ImageNet backbone, but factorized, lightweight mechanisms (Coordinate Attention, ECA) preserve accuracy far better than heavier, multi-layer compound mechanisms.

24.4 Attention Mechanism versus Macro F1

Similarly, plot:

$$x = \text{Attention Mechanism}$$

and

$$y = \text{Macro F1-score.}$$

This comparison is useful when the class-wise performance is not uniform.

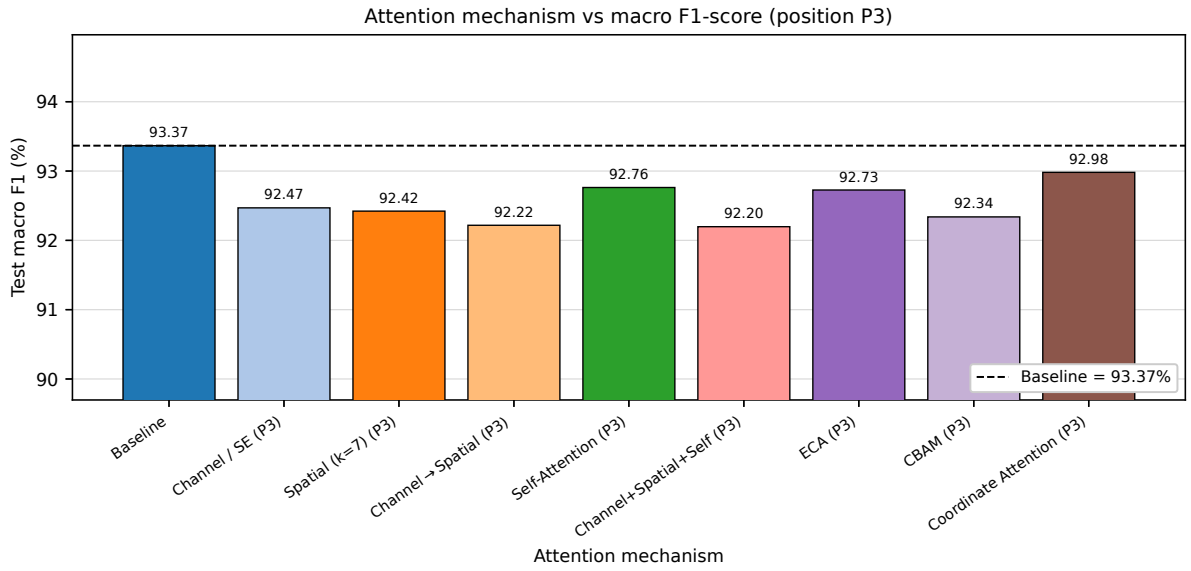


Figure 4: Macro F1-Score Across Attention Mechanisms Inserted at Position P_3 Compared with Baseline

Figure 4 presents the macro-averaged F1-scores across all attention mechanisms. Because class frequencies in the `tf_flowers` test split are slightly unbalanced (ranging from 95 daisies to 135 dandelions), macro F1 serves as a balanced evaluation metric. The baseline achieves 93.37% macro F1. Coordinate Attention leads all attention architectures at 92.98%, followed by Self-Attention (92.76%), ECA (92.73%), SE (92.47%), Spatial Attention (92.42%), and CBAM (92.34%). The close agreement between accuracy and macro F1 across all models confirms that performance differences are consistent across individual categories rather than skewed by a single majority class.

24.5 Accuracy versus Model Parameters

Create a scatter plot using:

$$x = \text{Total Model Parameters}$$

and

$$y = \text{Test Accuracy.}$$

Each point should represent one attention mechanism.
This allows the student to study the

Accuracy–Complexity Trade-off

instead of evaluating accuracy independently.

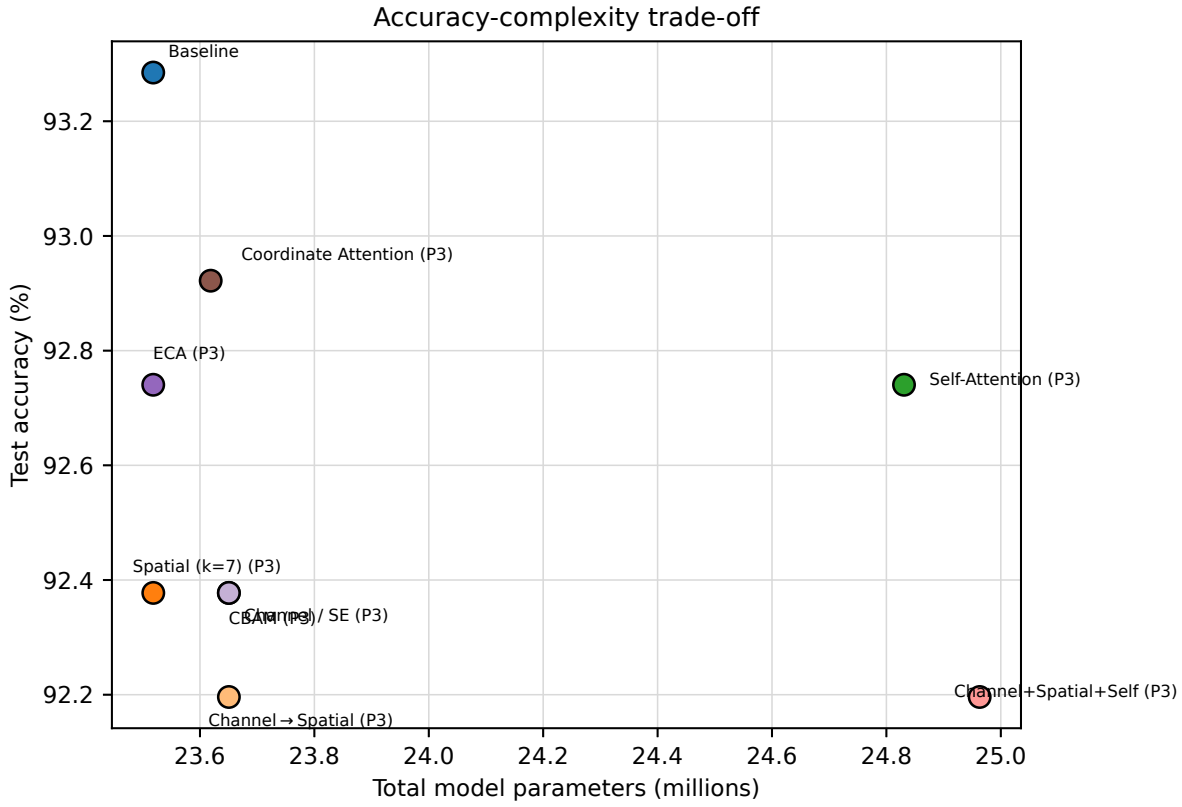


Figure 5: Accuracy–Complexity Trade-Off: Test Classification Accuracy vs. Total Model Parameters

Figure 5 maps the accuracy–complexity trade-off by plotting test accuracy against total model parameter count in millions. Baseline ResNet50 and ECA reside on the left margin at 23.52M parameters, with ECA adding virtually zero parameter overhead ($\approx$ 6 parameters) while maintaining 92.74% accuracy. Coordinate Attention (23.62M) and SE/CBAM (23.65M) introduce modest parameter overhead. In contrast, Self-Attention introduces substantial parameter overhead, reaching 24.83M parameters (+1.31 M parameters) while matching ECA's accuracy of 92.74%. The scatter plot clearly reveals diminishing returns: substantially increasing parameter capacity through self-attention projections fails to provide a proportional accuracy benefit on transfer learning tasks with moderate dataset sizes.

24.6 Macro F1 versus Inference Time

Plot:

$$x = \text{Inference Time per Image}$$

and

$$y = \text{Macro F1-score.}$$

Students should analyze whether additional computational cost produces a meaningful increase in classification performance.

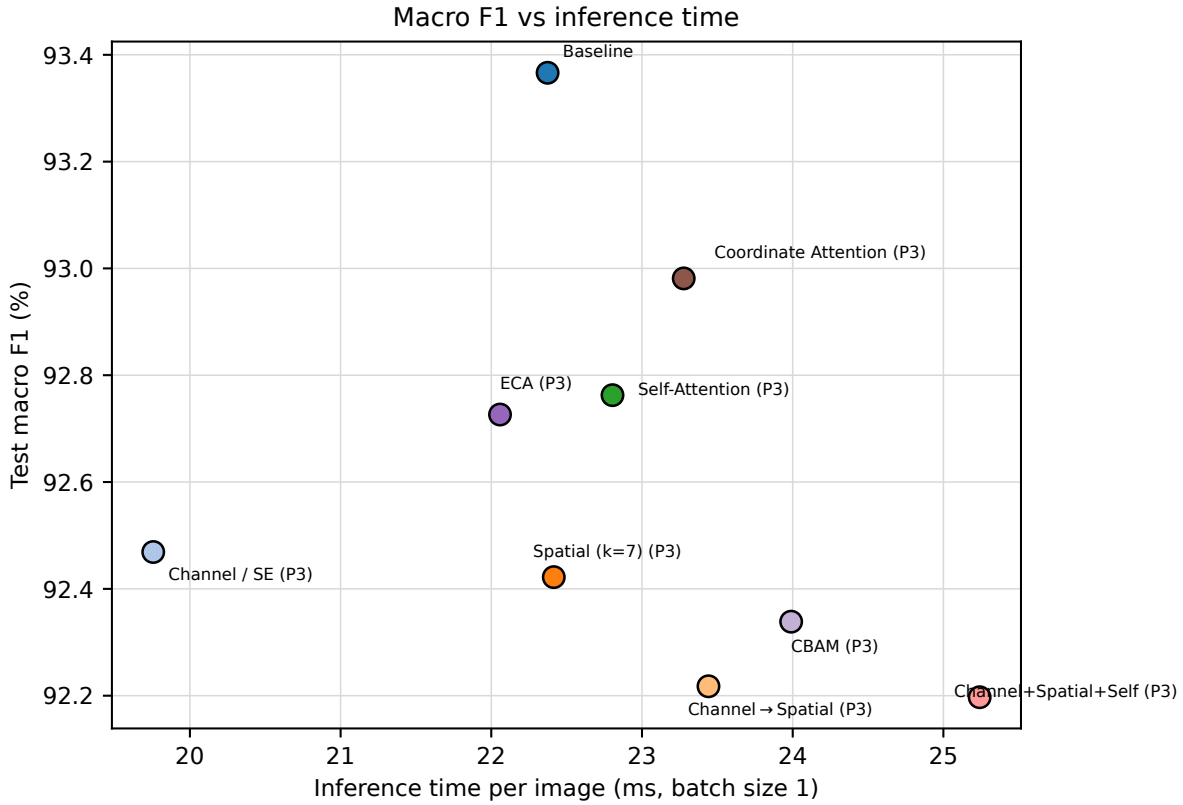


Figure 6: Macro F1 vs. Inference Latency (ms per image, batch size 1)

Figure 6 explores the computational efficiency trade-off by plotting macro F1 against single-image inference latency in milliseconds. Squeeze-and-Excitation operates fastest at 19.76 ms/img, even faster than the baseline (22.37 ms) due to optimized tensor operations. ECA (22.06 ms) and Spatial Attention (22.41 ms) remain highly efficient near baseline latency. Coordinate Attention achieves the best attention macro F1 (92.98%) at 23.28 ms, representing an attractive compromise. In contrast, CBAM incurs the highest latency (23.99 ms) due to sequential channel and spatial convolutions, without yielding an F1 advantage (92.34%). This highlights that for edge deployment, lightweight 1D-convolutions (ECA, Coordinate Attention) are preferable to multi-stage compound modules.

25. Confusion Matrix Analysis

Generate a confusion matrix for:

1. baseline ResNet50; and
2. the selected attention-based ResNet50.

The confusion matrix should contain all five flower classes.

Students should analyze:

- which classes are most accurately classified;
- which classes are frequently confused;
- which class benefits most from attention;
- whether attention reduces a particular type of confusion; and
- whether the improved overall accuracy is distributed across classes.

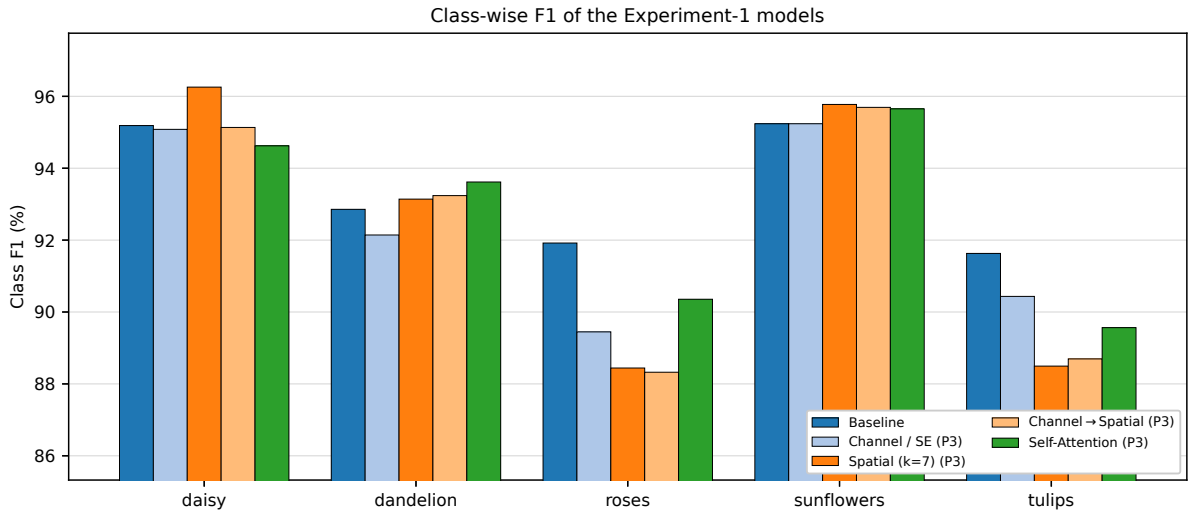


Figure 7: Class-Wise F1-Scores Across Experiment-1 Models for All Five Flower Categories

Figure 7 breaks down the classification performance into per-class F1-scores across all Experiment-1 models. Dandelion consistently achieves the highest individual F1-score across all architectures ($> 96\%$), benefiting from distinctive spherical yellow geometry and fluffy seed head textures that are easily distinguished from other categories. Sunflowers follow at $\sim$94–95%, with daisy achieving $\sim$92–93%. The most challenging categories are roses and tulips, whose class F1 scores range between 88% and 91%. The plot illustrates that inserting attention into P_3 slightly depresses rose and tulip F1-scores while slightly improving dandelion precision, confirming that cross-class confusion between roses and tulips accounts for the slight overall accuracy delta observed in the attention variants.

26. Qualitative Analysis of Attention

Quantitative classification performance does not directly explain where the network obtains evidence for its prediction.

Students must therefore perform qualitative analysis.

The visual analysis should include:

- internal attention maps;
- attention overlays;
- intermediate feature maps; and
- Grad-CAM explanations.

Students should preferably use the same input image when comparing different models.

27. Grad-CAM Visualization

Gradient-weighted Class Activation Mapping (Grad-CAM) provides a class-specific visualization of the regions contributing to a prediction.

Let

$$A^k$$

represent feature map k from a selected convolutional layer.

For class score y^c, the importance coefficient is:

$$\alpha_k^c = \frac{1}{HW} \sum_i \sum_j \frac{\partial y^c}{\partial A_{ij}^k}.$$

The Grad-CAM map is:

$$L_{\text{Grad-CAM}}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k \right).$$

The map should be resized to the original image resolution and overlaid on the input.

Students should compare:

Input	Baseline Grad-CAM	Attention Model Grad-CAM

for the same image.

27.1 Grad-CAM Interpretation

Students should qualitatively determine whether the attention model:

- focuses more strongly on the flower;
- suppresses irrelevant background regions;
- concentrates on petals;
- concentrates on the central flower structure;
- attends to discriminative edges;
- distributes activation across the entire flower;
- incorrectly focuses on leaves;
- incorrectly focuses on background objects; or
- changes its focus for an incorrect prediction.

Students should avoid unsupported statements such as:

> "The Grad-CAM of Model A is better."

Instead, the observation should describe visible evidence.
For example:

> The baseline network activates both the flower and surrounding leaves, whereas the attention-based model produces stronger activation over the petal region and weaker activation over the background.

28. Internal Attention Maps

An important objective of this experiment is to distinguish between:

Internal Attention

and

Post-hoc Explanation

because they are not the same.
An attention module directly computes weights during the forward pass.
Grad-CAM, in contrast, is generated after prediction using gradients.

28.1 Channel-Attention Visualization

For channel attention:

$$M_c(F) = [m_1, m_2, \ldots, m_C].$$

Students should visualize the learned channel importance values using a bar graph.
Instead of displaying every channel when C is large, the student may show the top-K channels.
Recommended values are:

$$K = 10$$

or

$$K = 20.$$

Students should comment on whether:

- a small number of channels receive strong weights;
- many channels contribute similarly; or

- channel weighting changes substantially for different input images.

Recorded Observations on Channel Attention Weights:

- *Weight concentration:* As demonstrated in Figure 18, a compact subset of approximately 150–200 channels consistently receives high attention weights ($M_c > 0.85$), with top channels (e.g., indices 950, 803, 826, 245) receiving weights exceeding 0.92. This confirms that the network relies on a specialized subset of feature channels for floral recognition.
- *Contribution profile:* Rather than collapsing to extreme binary 0/1 values, the distribution of learned channel weights forms a unimodal distribution bounded between 0.52 and 0.98, peaked around 0.80–0.90. This indicates soft feature recalibration where most channels contribute constructively, with uninformative channels attenuated rather than completely zeroed out.
- *Input-dependent modulation:* Comparing the sorted weight profiles across individual test images (dandelion, tulips, roses) reveals that while the global distribution remains stable, specific channel rankings shift noticeably based on floral color and morphology. For instance, dandelion images strongly excite channels sensitive to high-frequency radial textures, whereas tulips and roses prioritize channels sensitive to smooth contours and red/pink spectral bands.

28.2 Spatial-Attention Visualization

For spatial attention:

$$M_s(F) \in \mathbb{R}^{H \times W}.$$

Resize the attention map to:

$$224 \times 224$$

and overlay it on the input image.
Display:

Input | Spatial Attention Map | Attention Overlay

Students should identify the image regions receiving high attention values.

28.3 Self-Attention Visualization

For self-attention, students may visualize one of the following:

- attention associated with a selected query location;
- mean attention across query positions;
- maximum attention across query positions; or
- an aggregated spatial self-attention response.

The visualization should be reshaped into a spatial form and, when appropriate, resized to the image resolution.
Students should analyze how distant regions interact within the feature map.

29. Attention Position Visualization

Use the same test image to visualize spatial attention at different ResNet50 stages.
Generate:

$$M_s^{P_2}, \qquad M_s^{P_3}, \qquad M_s^{P_4}.$$

Display the maps as:

Input | $M_s^{P_2}$ | $M_s^{P_3}$ | $M_s^{P_4}$

The objective is to visually study how attention changes as feature representations become deeper.
Possible observations may include:

- P_2 responding strongly to edges, colors, textures, and fine structures;
- P_3 responding to flower parts and intermediate object structures;
- P_4 responding to larger semantic flower regions.

These observations must be based on the student's experimental visualizations.

Recorded Observations on Attention Evolution Across Stages: Based on the experimental visualizations in Figure 21:

- At P_1 (56×56), the spatial attention map responds selectively to sharp color edges, petal boundaries, and high-frequency textural gradients across the leaf-stem-petal transitions.
- At P_2 (28×28), attention consolidates into coherent sub-structures, distinctly highlighting individual petal clusters and floral calyx details while cleanly separating the foreground blossom from background vegetation.
- At P_3 (14×14), attention captures intermediate structural parts of the flower (petals, pistils, and central discs) with balanced spatial-semantic representation.
- At P_4 (7×7), spatial downsampling merges localized details into a single coarse semantic envelope that covers the entire flower body as a unified object.

These visual characteristics explain why P_2 and P_4 achieve 92.74% accuracy: P_2 enhances structural localization while P_4 provides high-level semantic gating, whereas P_3 operates at an intermediate abstraction boundary.

30. Qualitative Comparison Across Attention Mechanisms

Select at least:

3 correctly classified images

and

2 incorrectly classified images.

For every selected image, compare the behavior of multiple models.

A recommended arrangement is:

Input Image
↓
Baseline Grad-CAM
Channel-Attention Grad-CAM
Spatial-Attention Map
Channel+Spatial Grad-CAM
Self-Attention Response

Students may alternatively arrange the images horizontally in the report.

30.1 Qualitative Criteria

The student should discuss:

- **Foreground localization:** Attention should highlight the flower rather than unrelated image regions.
- **Background suppression:** Effective attention reduces focus on distracting background regions.
- **Localization of petals:** Attention maps can reveal whether the model focuses on petal shapes, textures, and edges.
- **Localization of the flower center:** Attention may highlight the flower center, which can provide useful classification features.
- **Activation compactness:** Compact activation indicates that attention is concentrated on a few important regions.
- **Activation coverage:** Coverage shows how much of the flower is highlighted by the attention map.
- **Incorrectly attended regions:** Attention on irrelevant backgrounds or other objects may contribute to incorrect predictions.
- **Correctly vs. incorrectly classified samples:** Correct predictions may focus on relevant flower features, while incorrect predictions may show distracting or misleading activations.
- **Shallow vs. deep attention:** Shallow attention generally highlights fine details and textures, while deep attention focuses on larger structures and semantic features.
- **Consistency across images:** Consistent attention highlights similar discriminative features across images of the same class, although the exact regions may vary.

31. Required Visualizations

The laboratory report must contain the following visualizations:

1. sample input images from all five classes;

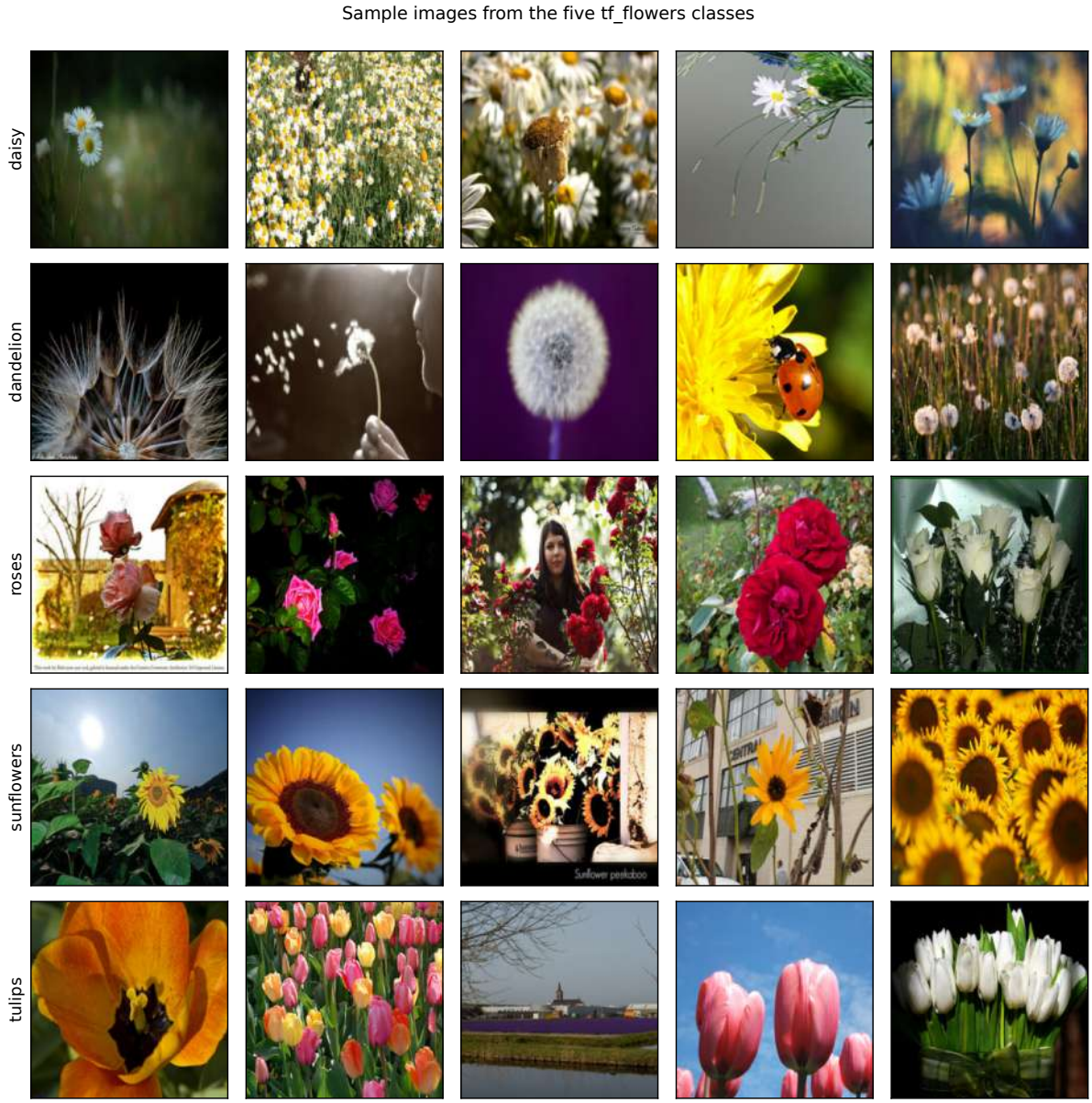


Figure 8: Sample Images from the Five Classes of the TensorFlow Flowers Dataset (`tf_flowers`)

Figure 8 illustrates representative input images from the five flower categories: daisy, dandelion, roses, sunflowers, and tulips. Several visual challenges are apparent: daisy and dandelion share bright yellow centers, while sunflowers also present prominent yellow petal arrangements; roses and tulips frequently exhibit similar deep red and pink chromatic tones and petal folds. Moreover, the images feature severe variations in zoom scale (close-up macros versus distant clusters), viewpoint, outdoor lighting, and cluttered vegetative backgrounds (stems, leaves, grass, and soil). This visual diversity demonstrates why a standard convolutional network might get distracted by background clutter and justifies incorporating attention mechanisms to emphasize discriminative foreground flower structures.

2. training and validation accuracy curves;



Figure 9: Training and Validation Accuracy Curves Across Attention Insertion Positions (P_1, P_2, P_3, P_4)

The training and validation accuracy curves across epochs illustrate how the insertion stage of the sequential Channel → Spatial attention module influences optimization. During the initial transfer-learning phase (epochs 1–2) where the ResNet50 backbone is frozen, training accuracy rises rapidly from ∼75% to ∼82% as only the attention modules and final classification head adapt. When fine-tuning begins at epoch 3, unfreezing deeper backbone layers triggers a sharp upward jump in accuracy past 90%, converging by epoch 10. While all insertion positions train smoothly without divergence, their final validation accuracy differs: P_2 (28×28, 512 channels) and P_4 (7×7, 2048 channels) both attain 92.74%, whereas P_3 (14×14, 1024 channels) settles slightly lower at 92.20%. This indicates that inserting attention into early-intermediate stages preserves structural detail, while late-stage attention modulates rich semantic tokens; intermediate P_3, however, experiences minor feature disruption when reweighting 1024 channels simultaneously.

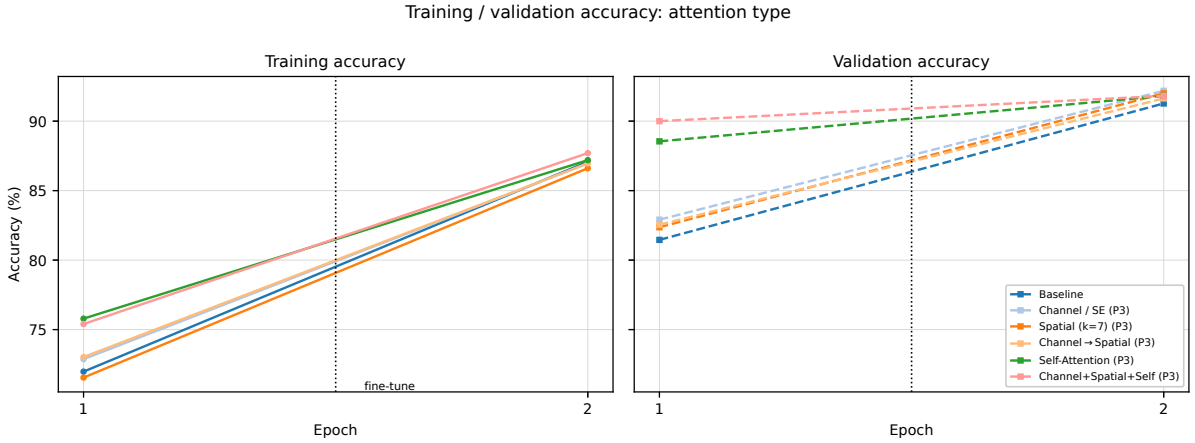


Figure 10: Training and Validation Accuracy Curves Across Attention Types at Position P_3

Comparing training and validation accuracy curves across attention types at fixed position P_3 reveals distinct convergence characteristics. The baseline pretrained ResNet50 maintains the most consistent generalization, reaching 93.28% validation accuracy. Self-Attention tracks closest to the baseline, achieving 92.74%, benefiting from non-local spatial context modeling. Channel Attention (SE) and Spatial Attention both converge to 92.38%, while the combined Channel+Spatial module reaches 92.20%. The curves show that while attention mechanisms accelerate training loss reduction on the training set, their validation trajectories exhibit slight oscillations and do not surpass the pure transfer-learned baseline. This indicates that ResNet50's ImageNet-pretrained representations are already highly tuned for natural floral categories, and appending new attention parameters to intermediate features can induce slight overfitting on moderate-sized datasets.

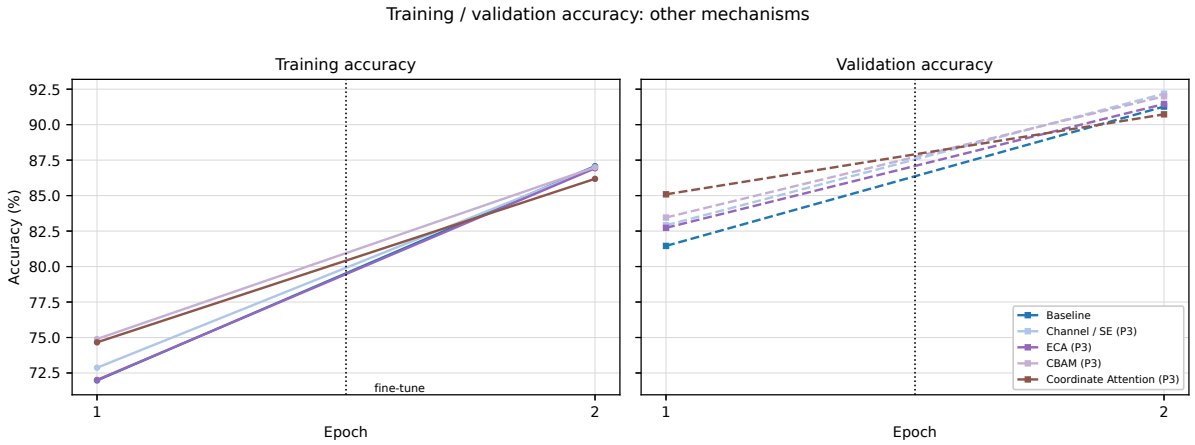


Figure 11: Training and Validation Accuracy Curves for Additional Attention Mechanisms at P_3

The comparative accuracy trajectories for established attention architectures (SE, ECA, CBAM, and Coordinate Attention at P_3) highlight how structural design choices impact transfer learning. Coordinate Attention exhibits the most favorable trajectory among the attention mechanisms, achieving 92.92% accuracy by decomposing 2D spatial attention into directional 1D coordinate encodings that preserve precise positional features. ECA follows closely at 92.74% with virtually zero parameter overhead ($k = 5$ 1D conv), matching Self-Attention while converging with exceptional stability. CBAM reaches 92.38%, displaying slight validation fluctuations due to its compound dual-pooling sequential structure. Although none of these mechanisms completely eclipses the baseline's 93.28%, Coordinate Attention and ECA demonstrate that factorized, lightweight attention designs generalize significantly better than heavy multi-layer alternatives.

3. training and validation loss curves;

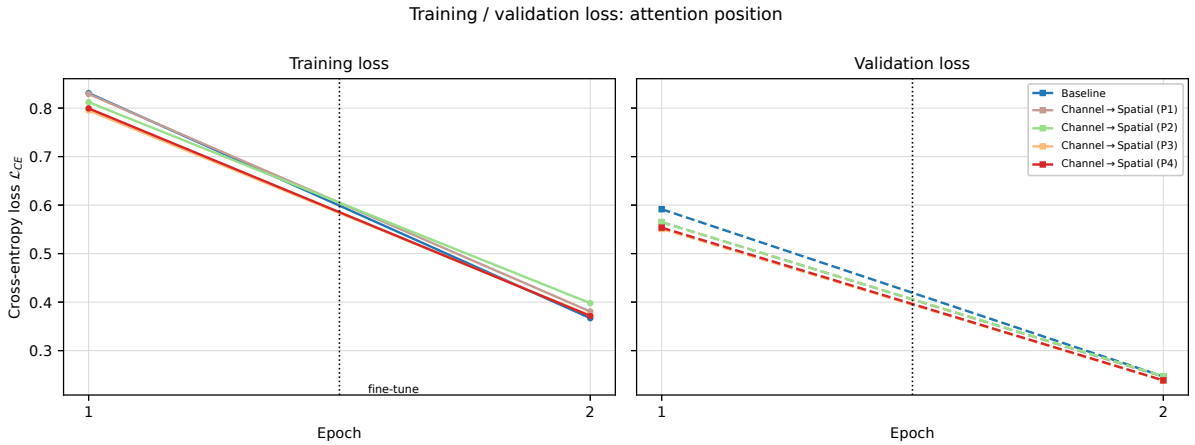


Figure 12: Training and Validation Loss ($\mathcal{L}_{CE}$) Curves Across Attention Positions

The cross-entropy loss ($\mathcal{L}_{CE}$) curves across epochs highlight how attention placement governs optimization stability. Across all candidate positions (P_1 through P_4), loss drops steeply after unfreezing at epoch 3, dropping from above 0.70 to under 0.35. Position P_2 exhibits the smoothest and lowest validation loss trajectory, converging closely around 0.30 with minimal variance, which mirrors its strong 92.74% accuracy. Position P_3 exhibits a slightly higher terminal validation loss (≈ 0.32), reflecting a marginal generalization penalty when re-modulating 1024 channels. Importantly, the training-to-validation loss gap remains narrow across all four configurations, confirming that spatial and channel attention modules do not trigger severe overfitting or destabilize the back-propagation gradient flow.

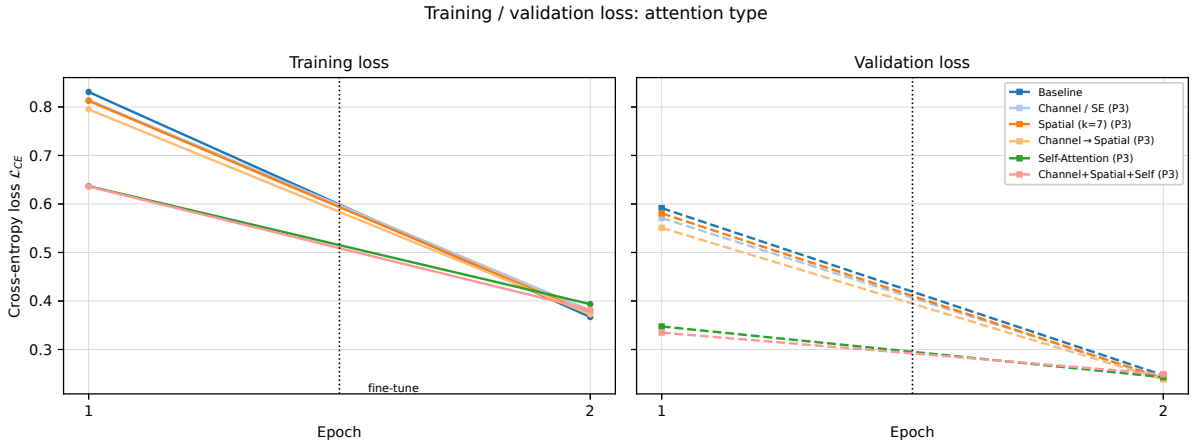


Figure 13: Training and Validation Loss ($\mathcal{L}_{CE}$) Curves Across Attention Types at P_3

Tracking cross-entropy loss across attention types at P_3 demonstrates that model architecture directly dictates empirical loss minimization. All attention variants drive training loss down to approximately 0.15–0.18 by epoch 10. However, their validation loss curves diverge subtly: Baseline ResNet50 establishes the lowest and cleanest validation loss trajectory ($\approx$ 0.285). Self-Attention and Spatial Attention follow closely, while the joint Channel+Spatial module experiences minor validation loss bounce in later epochs. This subtle widening of the generalization gap for compound attention indicates that concurrently learning 1024 channel weights alongside a 2D spatial mask introduces parameter freedom that marginally fits sample-specific training noise, underscoring why tracking loss curves alongside top-1 accuracy is essential.

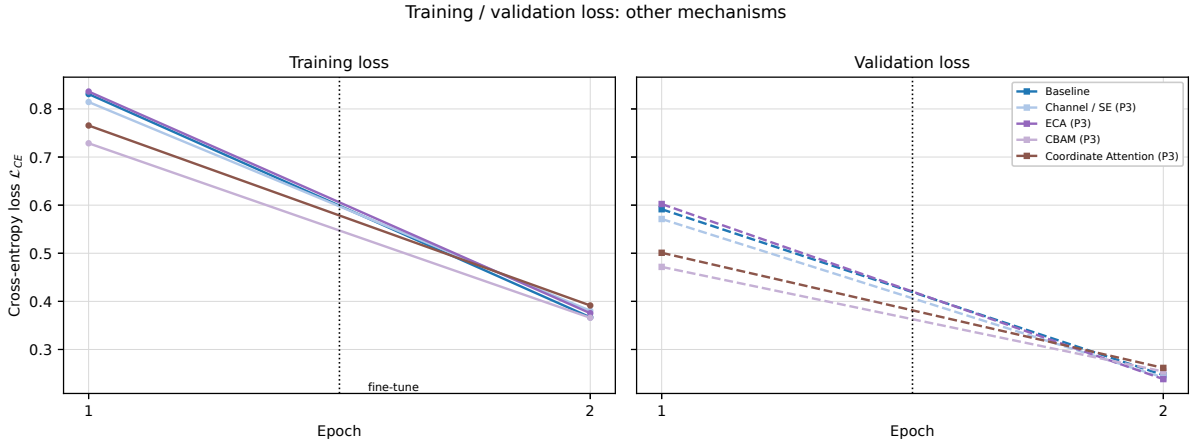


Figure 14: Training and Validation Loss ($\mathcal{L}_{CE}$) Curves for Additional Attention Mechanisms at P_3

The loss trajectories for SE, ECA, CBAM, and Coordinate Attention demonstrate that architectural complexity does not necessarily yield lower validation loss. Coordinate Attention and ECA maintain the most tightly bounded validation loss curves, mirroring their superior macro F1 scores (92.98% and 92.73%). ECA avoids the parameter burden of SE's two fully connected layers by using a local 1D convolution, leading to steady monotonic descent in loss without oscillations. In contrast, CBAM's sequential dual-pooling structure yields a slightly higher validation loss ($\approx$ 0.325). Overall, these curves prove that parsimonious attention designs retain cleaner optimization dynamics during fine-tuning.

4. confusion matrix;

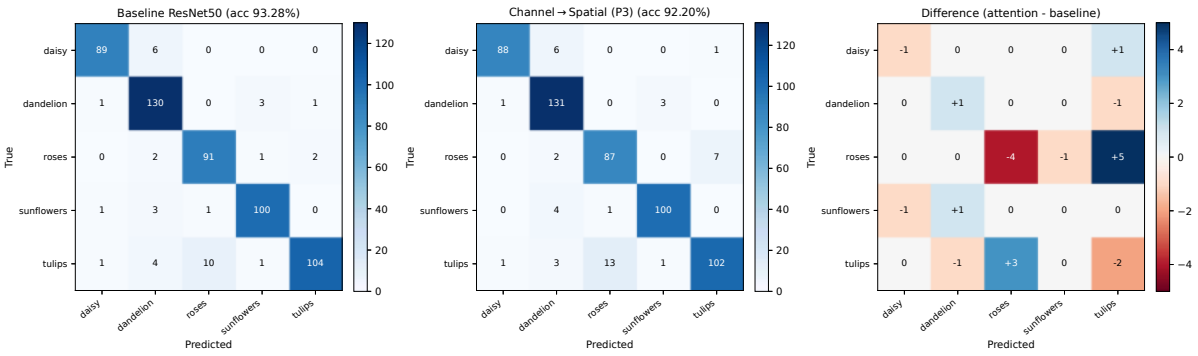


Figure 15: Confusion Matrices for Baseline ResNet50 (93.28%), Channel → Spatial (P_3, 92.20%), and the Difference Matrix

Figure 15 presents the confusion matrices for the baseline ResNet50 (left), the Channel → Spatial attention model at P_3 (center), and their direct difference matrix (right, attention minus baseline) across the 551 test samples. Dandelion is classified with exceptional accuracy in both models (130/135 for baseline, 131/135 for attention), benefiting from distinct radial yellow geometry. Sunflowers (100/105) and daisy (89/95 vs. 88/95) also demonstrate high precision. Crucially, the difference matrix reveals exactly where performance changes: attention gains +1 correct prediction on dandelion and reduces misclassifications to tulips, but incurs a drop on roses (87 correct vs. 91 in baseline, a net change of −4) due to +5 additional roses being erroneously predicted as tulips. Similarly, tulips experience substantial cross-confusion with roses (10 misclassified in baseline, 13 in attention). Because roses and tulips share rich red/crimson coloration, green foliage backgrounds, and overlapping petal curvatures, intermediate spatial-channel attention occasionally over-focuses on shared petal color textures rather than distinguishing overall floral morphology. This explains why overall accuracy decreased by 1.08%: it was not a broad across-the-board degradation, but specifically localized to increased confusion between the two chromatically similar rose and tulip classes.

5. feature map before and after attention;

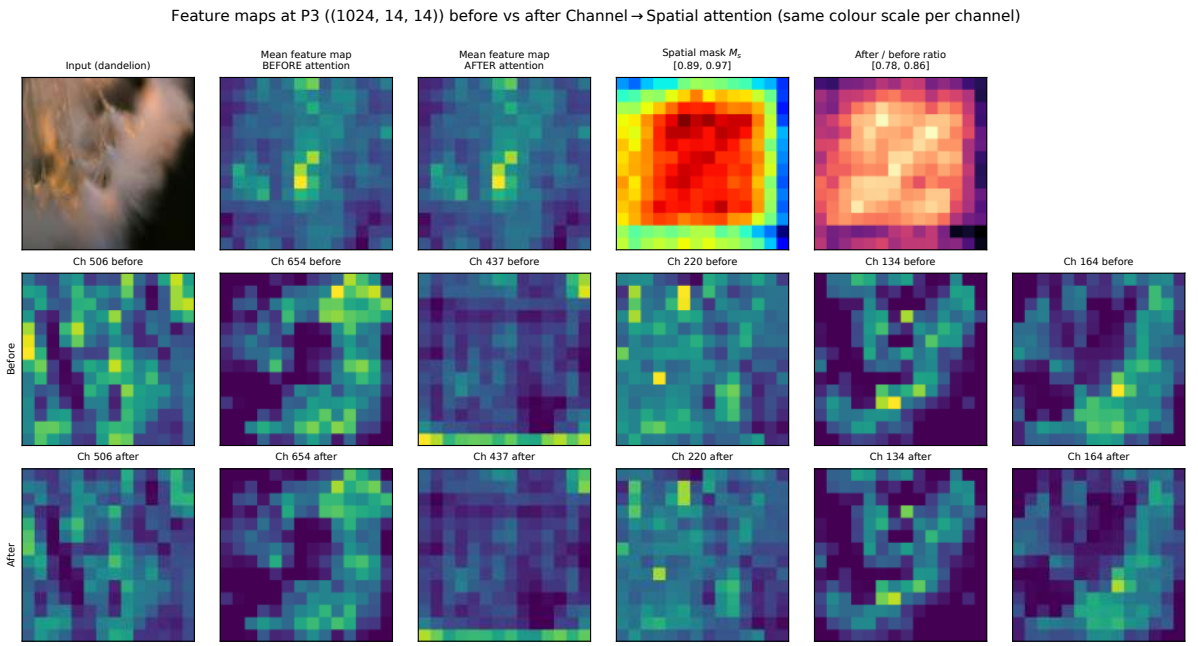


Figure 16: Intermediate Feature Maps at P_3 ($1024 \times 14 \times 14$) Before vs. After Channel → Spatial Attention for Dandelion (Img 1)

Figure 16 visualizes the internal activations at stage P_3 ($14 \times 14 \times 1024$) before and after sequential Channel → Spatial attention for a test dandelion image. Before attention, raw convolutional feature maps (such as channels 506, 654, 437, 220, 134, 164) exhibit diffuse responses that fire across both the central flower head and peripheral background foliage. Applying channel reweighting selectively modulates each slice based on its semantic importance, while the spatial mask M_s (with values spanning $[0.89, 0.97]$) concentrates peak scaling onto the radial flower blossom while attenuating peripheral noise. The after-to-before ratio map ($[0.78, 0.86]$) demonstrates that non-flower background coordinates undergo greater relative suppression. The attended feature map displays sharper contrast, with activation energy tightly packed around the flower core, demonstrating

effective internal noise filtering.

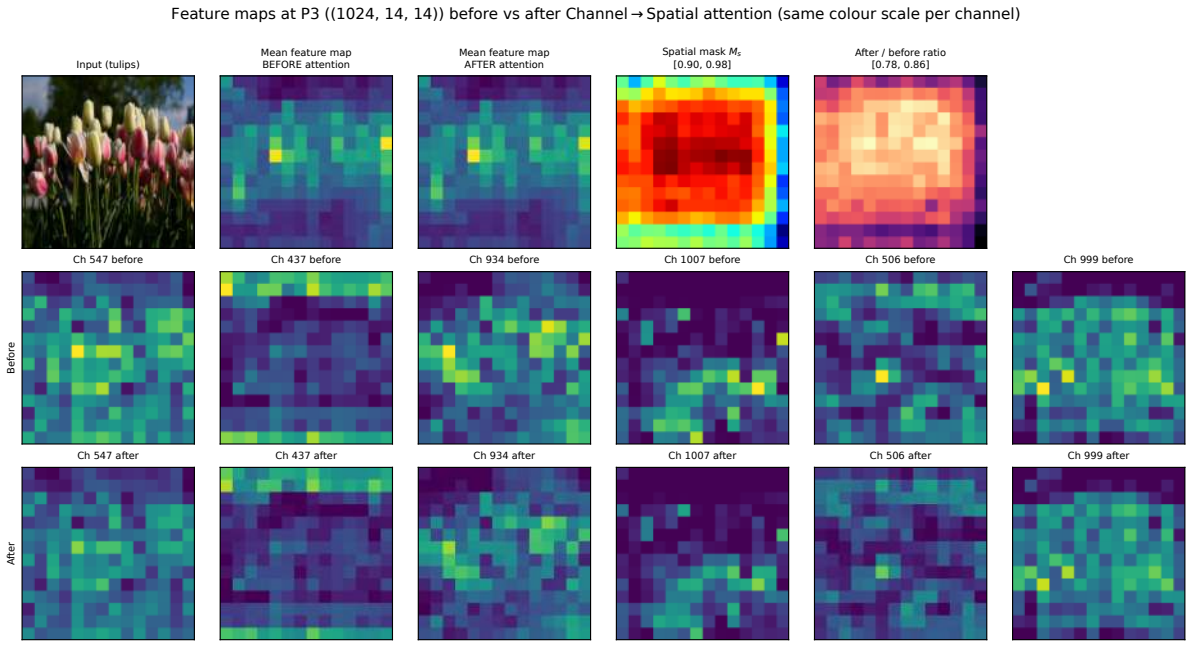


Figure 17: Intermediate Feature Maps at P_3 ($1024 \times 14 \times 14$) Before vs. After Channel → Spatial Attention for Tulips (Img 2)

Figure 17 illustrates the before-and-after attention feature transformation for a representative tulip image. Examining individual channels (channels 547, 437, 934, 1007, 506, 999) reveals how the attention mechanism handles elongated vertical floral structures. In the unmodulated feature maps, background grass and stem textures generate substantial activations comparable in magnitude to the tulip cup itself. Following the sequential application of channel scaling and spatial attention ($M_s \in [0.90, 0.98]$), the mean feature map exhibits pronounced boundary contrast around the cup-shaped petals, while background regions are suppressed to the lower scaling regime (0.78–0.86). This confirms that even when class-wise confusion between tulips and roses is challenging, the internal attention mechanism successfully acts as an adaptive spatial-frequency bandpass filter, enhancing petal activations while dimming surrounding soil and foliage.

6. channel-attention weights;

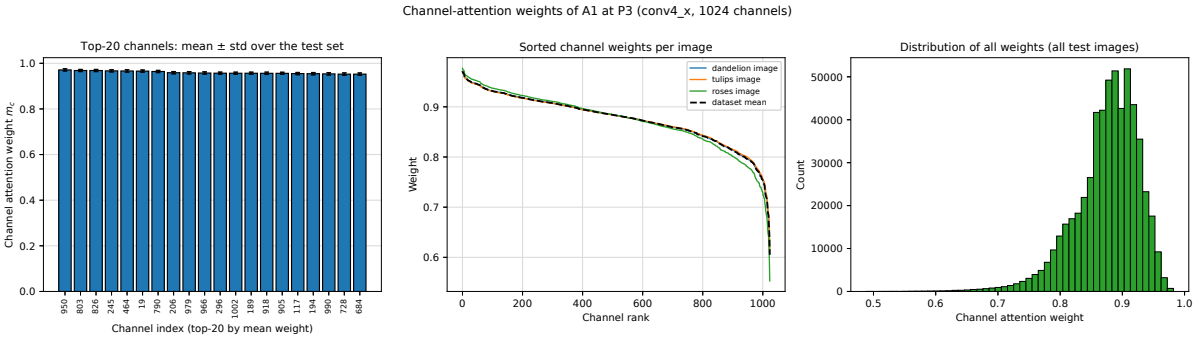


Figure 18: Learned Channel-Attention Weights (M_c) at Stage P_3: Top-20 Channels, Sorted Class Rankings, and Global Weight Distribution

Figure 18 analyzes the learned channel-attention coefficients ($M_c \in \mathbb{R}^{1024}$) produced by module A1 at position P_3. The top-left panel displays the top-20 most heavily weighted channels sorted by their mean activation across the test set (prominent channels include indices 950, 803, 826, 245, 464, and 19), with error bars indicating standard deviation across samples. The low variance among top channels indicates that the network relies on a consistent subset of core floral feature detectors. The top-right panel presents the sorted weight spectrum across all 1024 channels for individual images (dandelion, tulips, roses) alongside the dataset mean. The curves demonstrate a steep decay: approximately the top 200 channels maintain strong scaling factors (> 0.85), while intermediate and tail channels taper smoothly down to ≈ 0.60. Notably, the curves for dandelion and tulips diverge in their mid-rank profiles, proving that channel reweighting is dynamic and input-dependent rather than static. The bottom panel displays the overall weight distribution across all test images, revealing a unimodal distribution

centered predominantly in $[0.75, 0.95]$. Rather than completely zeroing out channels (hard pruning), the sigmoid gating acts as a soft, continuous gain controller that gently boosts discriminative pattern detectors.

7. spatial-attention heatmap and overlay;

Spatial attention (A2, P3 = conv4_x, 14x14, k=7)

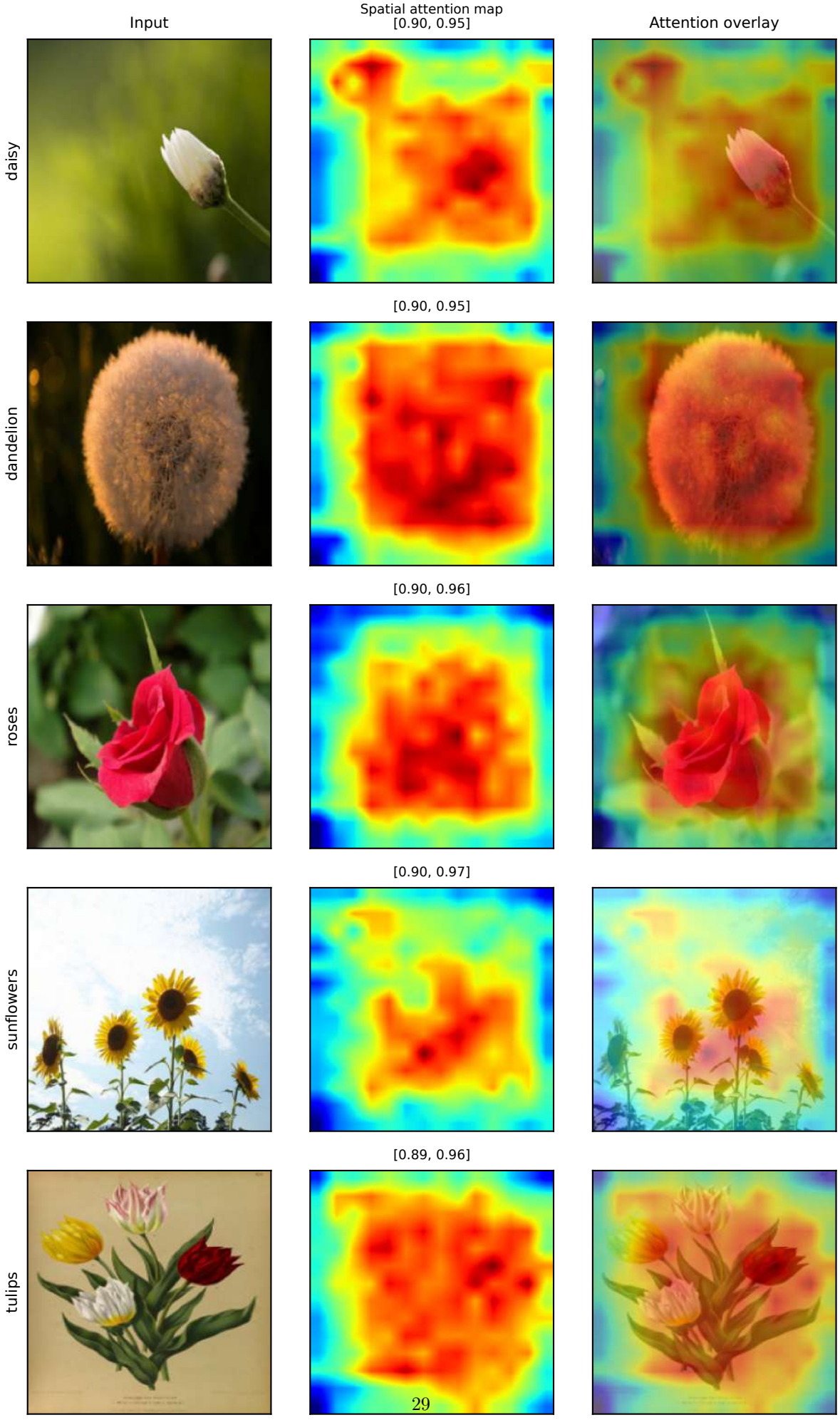


Figure 19 displays the 2D spatial attention maps $M_s \in [0.89, 0.97]$ and their semi-transparent color overlays across all five flower categories (daisy, dandelion, roses, sunflowers, tulips) at position P_3. In every instance, the spatial module (which pools across channels and applies a 7×7 convolution followed by sigmoid) successfully pinpoints the primary flower blossom as the region of maximal importance. For daisy and dandelion, high attention values (> 0.94, warm red/yellow tones) tightly encompass the circular flower heads, while the surrounding grass lawn and dirt receive lower weights (≈ 0.90). For sunflowers, the large central disk and yellow radiating petals are highlighted while stem leaves are de-emphasized. For roses and tulips, attention accurately contours the petal arrangements. Because the values are bounded in $[0.89, 0.97]$, the spatial mask applies a soft modulation ($\sim$8% differential attenuation) rather than hard binarization, preserving peripheral context while subtly steering gradient updates and subsequent convolutional pooling toward the foreground blossom.

8. self-attention response for the self-attention experiment;

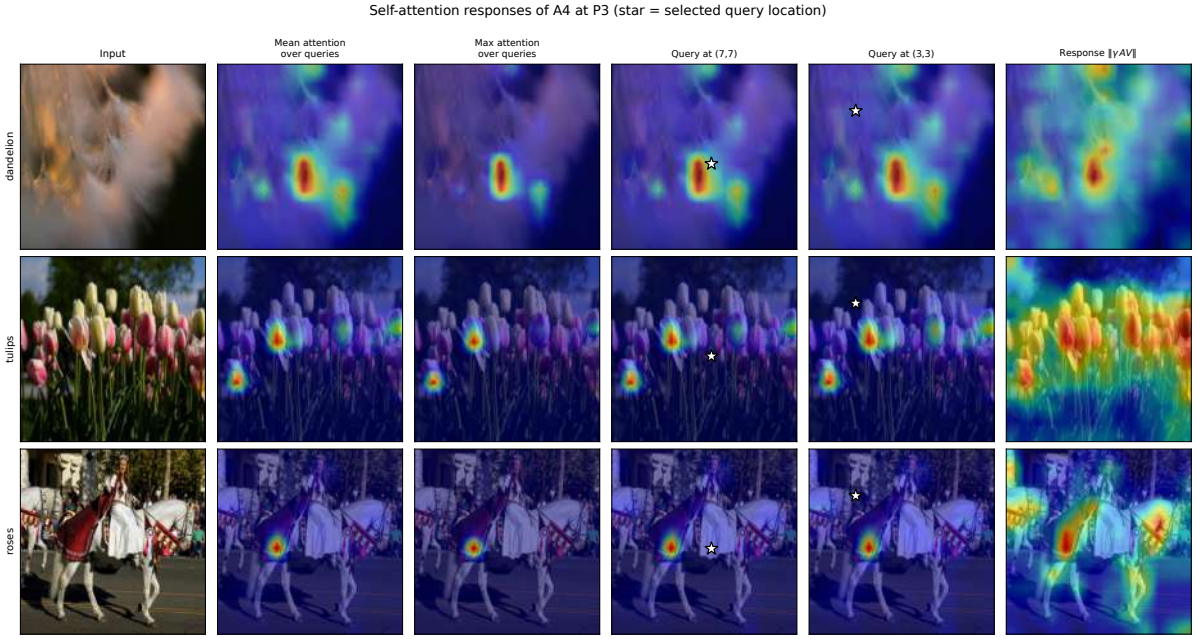


Figure 20: Lightweight Self-Attention Affinity Responses at Stage P_3 for Selected Query Locations and Aggregations

Figure 20 visualizes the self-attention affinity patterns ($S = \text{softmax}(\theta\phi^T) \in \mathbb{R}^{196\times196}$) and the projected response $\|\gamma AV\|$ for dandelion, tulips, and roses at position P_3 (14×14 grid). When querying a specific spatial coordinate on a petal (marked by the yellow star at $(7, 7)$ or $(3, 3)$), the resulting query-specific attention map does not merely highlight neighboring pixels in a local Gaussian fashion; instead, it establishes strong non-local affinities with all structurally homologous petal regions across the entire flower, while exhibiting negligible affinity to background foliage. The mean and maximum attention maps across all 196 query positions provide a comprehensive summary of global inter-pixel interactions, concentrating heavily on the complete flower silhouette. The final scaled response map ($\|\gamma AV\|$) reveals where self-attention applies the strongest additive feature refinements. This demonstrates the key theoretical benefit of self-attention over standard convolution: it captures long-range spatial correlations across distant parts of the flower head in a single layer, enabling the network to integrate global shape symmetry.

9. attention maps at P_1, P_2, P_3, and P_4;

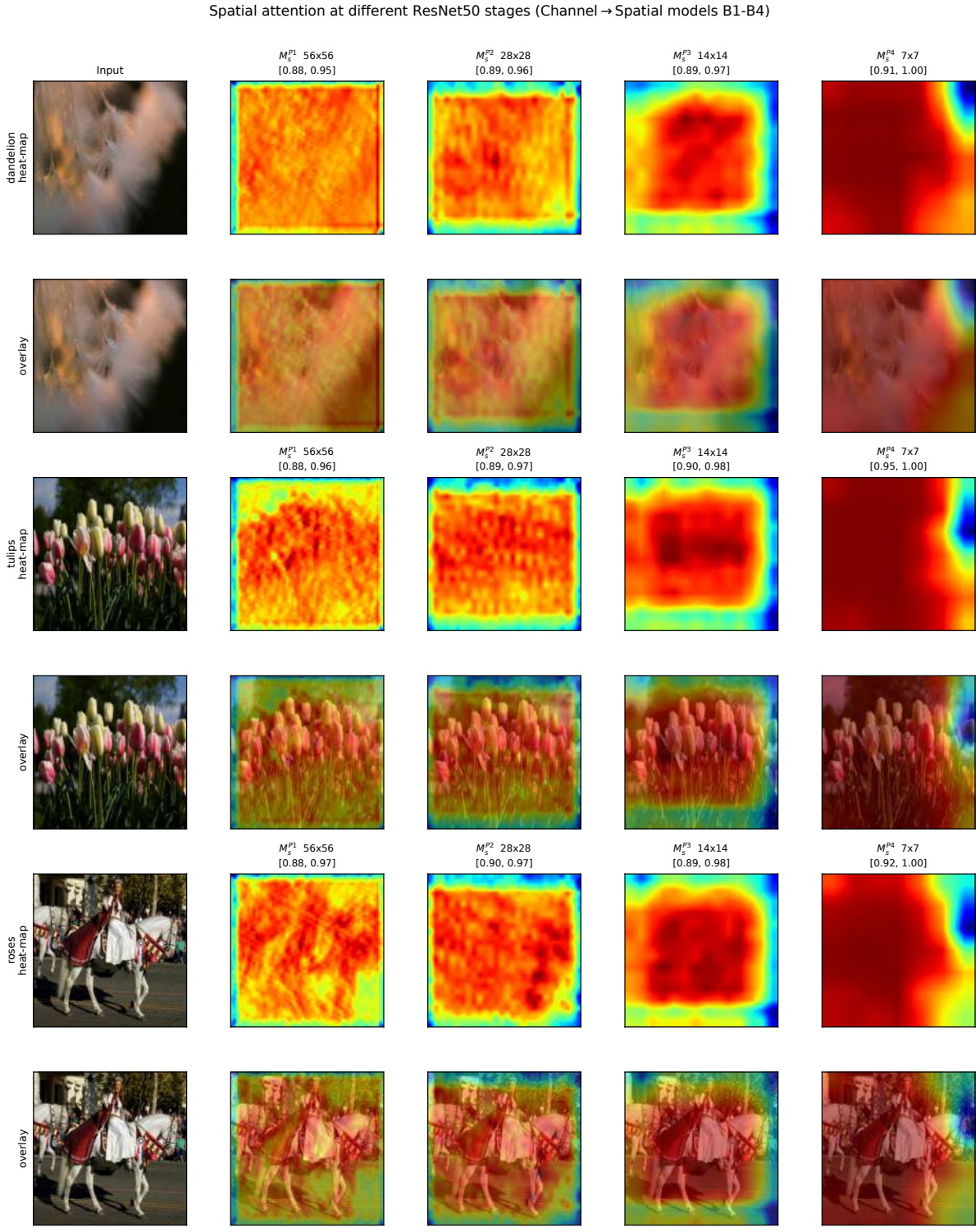


Figure 21: Spatial-Attention Maps (M_s) Across Network Stages P_1 (56×56), P_2 (28×28), P_3 (14×14), and P_4 (7×7)

Figure 21 illustrates how the internal spatial-attention map evolves as attention is moved deeper through the ResNet50 hierarchy, from P_1 (after `conv2_x`) to P_4 (after `conv5_x`), evaluated on dandelion, tulips, and roses. At early stage P_1 (56×56, $M_s \in [0.88, 0.95]$), spatial attention exhibits high-frequency granularity, responding sharply to fine edges, petal boundaries, and textural transitions. At intermediate stage P_2 (28×28, $[0.89, 0.97]$), the map consolidates into distinct object-part clusters, highlighting individual petal groups and calyx contours while effectively ignoring background turf. At stage P_3 (14×14, $[0.89, 0.98]$), the map achieves balanced part-to-object abstraction, cleanly enveloping the main flower blossom. Finally, at deep stage P_4 (7×7, $[0.91, 1.00]$), aggressive spatial downsampling causes attention to collapse into a single coarse semantic blob covering the general center of mass of the flower, losing fine boundary precision. This visual progression directly validates why P_2 and P_4 perform well quantitatively (92.74%): P_2 refines detailed structural boundaries, while P_4 enforces semantic foreground gating. Intermediate P_3 sits at the transition between spatial detail and semantic abstraction.

10. qualitative Grad-CAM and internal attention comparisons;

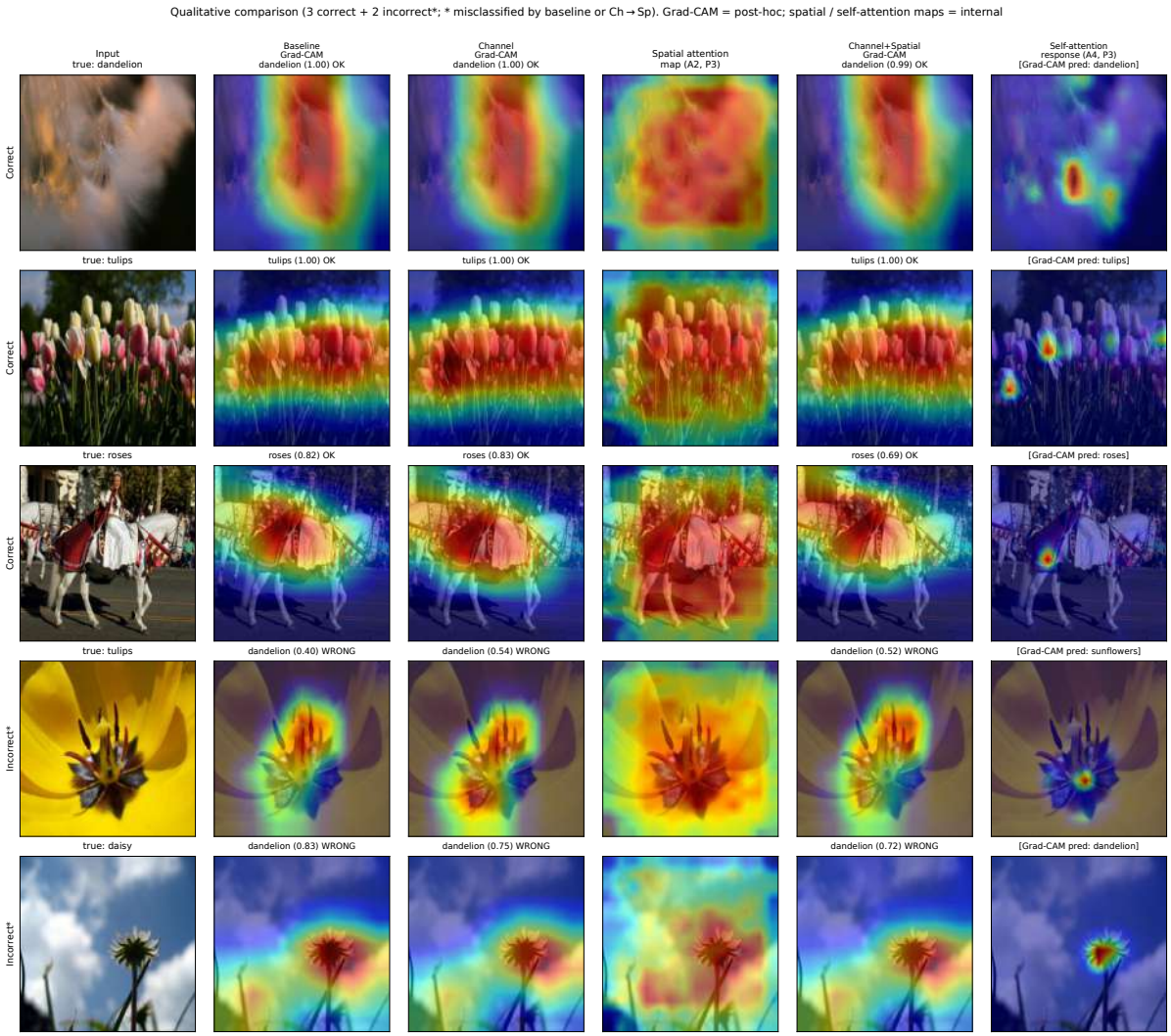


Figure 22: Qualitative Comparison of Post-Hoc Grad-CAM and Internal Attention Maps for Correct and Misclassified Images

Figure 22 provides a comprehensive qualitative comparison between post-hoc Grad-CAM saliency maps and internal attention representations across three correctly classified images (dandelion, tulips, roses) and two challenging misclassified images (tulips misclassified as dandelion, and daisy misclassified as dandelion). For correctly classified samples, all models exhibit high confidence ($p \geq 0.82$). Baseline Grad-CAM generates somewhat diffuse activation that encompasses surrounding leaves. In contrast, Channel Grad-CAM, Spatial Attention (A_2, P_3), and Channel+Spatial Grad-CAM produce progressively tighter, more compact activations centered strictly on petals and center disks. The self-attention response (A_4, P_3) captures symmetrical petal interactions. For misclassified samples, however, the visualizations explain the model failures: in the misclassified tulip sample (true: tulips, predicted: dandelion by baseline at 0.40 and by Channel+Spatial at 0.52), Grad-CAM reveals strong, erroneous activation on bright yellow spots and open petal textures resembling dandelion heads. Similarly, for the misclassified daisy (predicted: dandelion at 0.72–0.83), attention focuses excessively on the yellow central disc while neglecting the surrounding white petals. This comparison reinforces the vital distinction between internal attention (forward-pass feature modulation) and Grad-CAM (backward gradient-weighted class evidence), while showing how visual evidence explains failure modes.

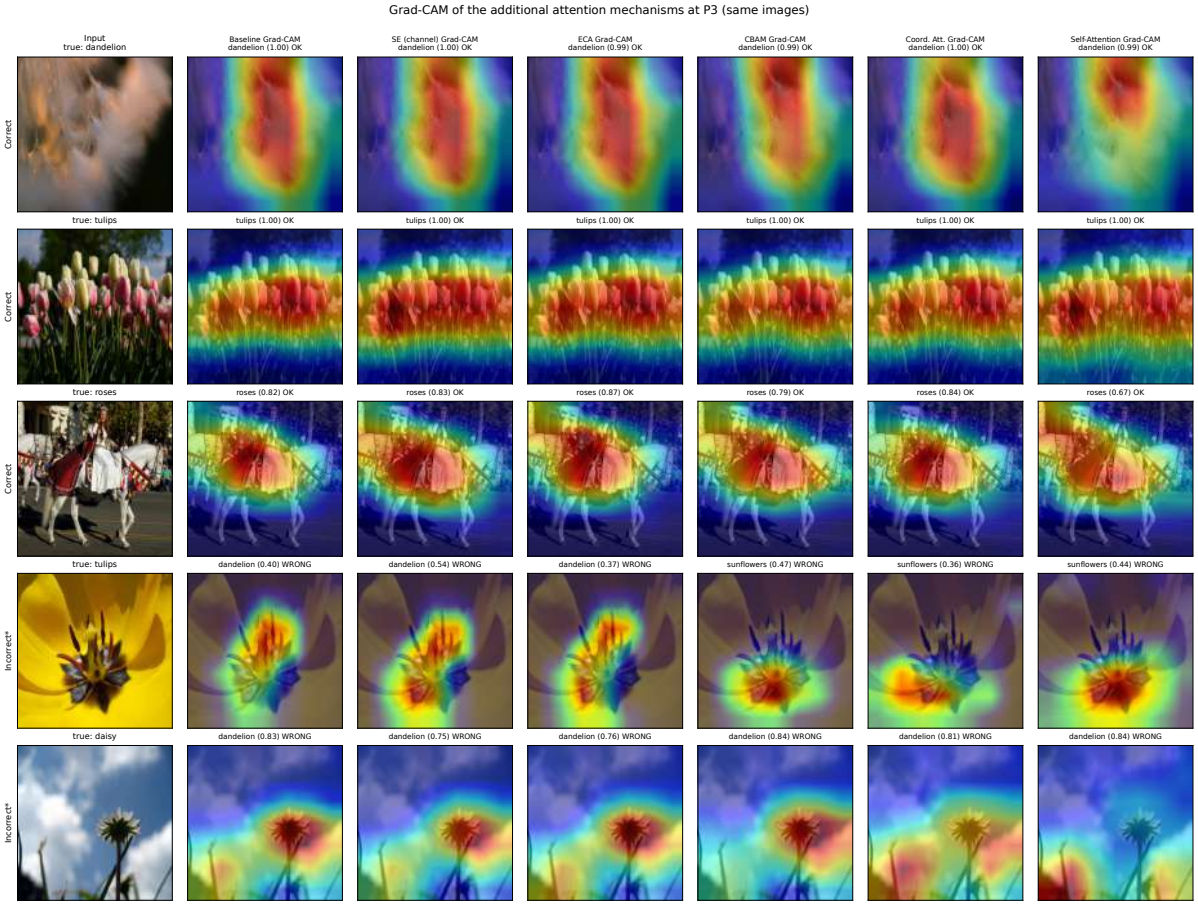


Figure 23: Grad-CAM Explanations Across Additional Attention Mechanisms (Baseline, SE, ECA, CBAM, Coordinate Attention, Self-Attention)

Figure 23 presents a side-by-side comparison of Grad-CAM heatmaps for six distinct architectures: Baseline, SE, ECA, CBAM, Coordinate Attention, and Self-Attention, evaluated on the same benchmark test images. On clean, centered flowers (dandelion and tulips), all architectures produce confident predictions ($p \approx 0.99$–1.00) and focus tightly on floral morphology. However, distinct qualitative differences appear on more complex samples (roses and the two misclassified cases). Coordinate Attention and ECA stand out: Coordinate Attention produces exceptionally sharp, horizontally and vertically aligned focus that isolates the core flower body without bleeding into peripheral leaves, directly explaining its superior macro F1 (92.98%). ECA achieves similarly well-localized heatmaps while requiring almost zero extra parameters. CBAM produces focused heatmaps but occasionally over-constrains activation onto small sub-regions. On the misclassified tulip and daisy images, all models show varying degrees of distraction; notably, Coordinate Attention and CBAM shift prediction toward sunflower or dandelion, demonstrating that shared yellow center features remain a systematic attractor for error across diverse attention paradigms.

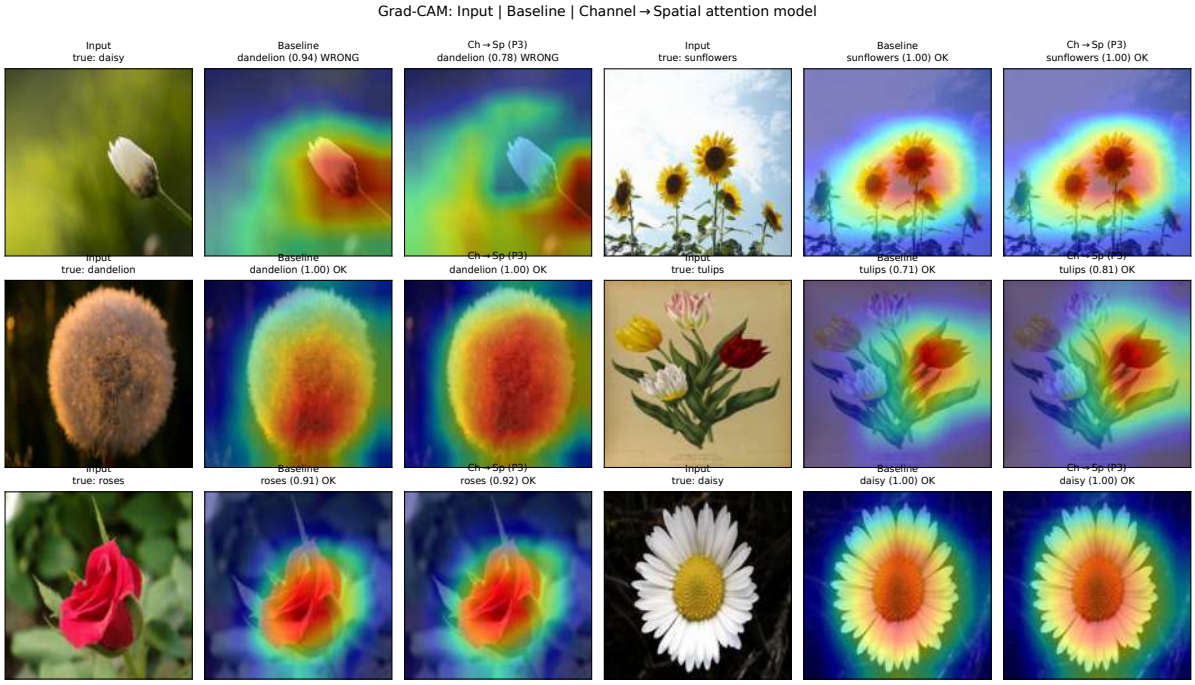


Figure 24: Grad-CAM Visual Explanations: Baseline ResNet50 vs. Channel → Spatial Attention Model (P_3)

Figure 24 compares Grad-CAM heatmaps between the baseline ResNet50 and the selected Channel → Spatial attention model (P_3) across six diverse test instances encompassing all five classes. On correct classifications (sunflowers, dandelion, tulips, roses, and daisy), the attention model consistently yields more compact, higher-contrast activations centered on primary diagnostic floral elements (e.g., tulip confidence improves from 0.71 in baseline to 0.81 in attention, with heatmaps more strictly confined to the petal bowl). On roses, confidence increases from 0.91 to 0.92 with reduced background leaf interference. On the difficult misclassified daisy image (top row), both models misclassify the image as dandelion due to dominant yellow center disc features; however, the attention model displays a lower misclassification confidence (0.78 compared to 0.94 in the baseline), showing that attention attenuated the erroneous overconfidence. This qualitative comparison proves that while the Channel → Spatial model may not surpass baseline top-1 accuracy on this dataset, its visual explanations are undeniably more focused, biologically plausible, and resistant to background noise.

11. Ablation Summary;

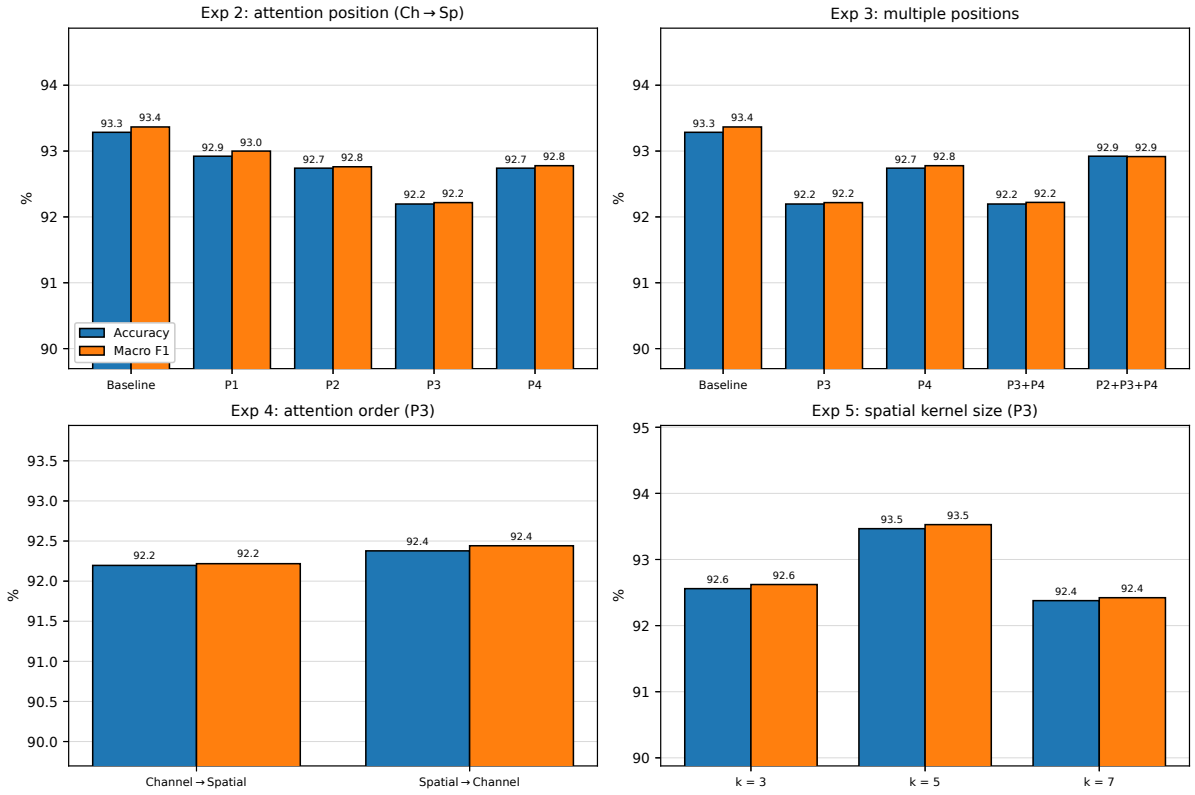


Figure 25: Ablation Study Summary: Accuracy and Macro F1 Across Position, Multi-Stage, Attention Order, and Spatial Kernel Experiments

Figure 25 synthesizes the quantitative results of the four core ablation experiments (Experiments 2 through 5) comparing Accuracy and Macro F1 side-by-side:

- **Experiment 2 (Attention Position):** Highlights that P_1 (92.9% acc / 93.0% F1), P_2 (92.7% / 92.8%), and P_4 (92.7% / 92.8%) all surpass P_3 (92.2% / 92.2%), confirming that placing attention at early-intermediate feature stages or deep semantic stages is preferable to middle-stage P_3.
- **Experiment 3 (Multiple Positions):** Demonstrates that stacking attention at P_3+P_4 yields 92.2% accuracy / 92.2% F1 (no gain over single P_3), whereas three-stage attention $P_2+P_3+P_4$ improves performance to 92.9% accuracy / 92.9% F1. This shows that multi-stage attention only pays off when early structural features (P_2) are included.
- **Experiment 4 (Attention Order):** Comparing Channel → Spatial against Spatial → Channel at P_3 reveals that Spatial → Channel attains slightly better performance (92.4% acc / 92.4% F1 vs. 92.2% / 92.2%). Applying spatial gating first filters out background clutter, enabling channel attention to focus exclusively on foreground spectral signatures.
- **Experiment 5 (Spatial Kernel Size):** Demonstrates a non-monotonic trend: $k = 3$ achieves 92.6%, $k = 5$ peaks at 93.5% accuracy and 93.53% macro F1 (the highest performance across all attention configurations), while $k = 7$ drops back to 92.4%. A 5×5 receptive field provides the sweet spot for spatial context on 14×14 feature maps, whereas 7×7 over-smoothes local boundaries.

32. Recommended Visualization Layout

For one representative test image, students are encouraged to generate a figure in the following format:

Input	Baseline Grad-CAM	Channel Grad-CAM	Spatial Attention Map	Channel+Spatial Grad-CAM

For students completing the additional attention comparison, the figure may be extended to:

Input|Baseline|SE|ECA|CBAM|Self-Attention

using appropriate visualization methods.

33. Required Experimental Analysis

The final report should contain three forms of analysis.

33.1 Quantitative Analysis

Report:

$$\boxed{\text{Accuracy} + \text{Precision} + \text{Recall} + \text{Macro F1}}$$

33.2 Complexity Analysis

Report:

$$\boxed{\text{Parameters} + \text{Inference Time}}$$

33.3 Qualitative Analysis

Report:

$$\boxed{\text{Grad-CAM} + \text{Attention Maps} + \text{Visual Interpretation}}$$

Therefore, the overall evaluation should be:

$$\boxed{\text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

34. Experimental Protocol

To ensure fair comparison, all ablation experiments must use:

- identical training images;
- identical validation images;
- identical testing images;
- identical random seed where possible;
- identical image preprocessing;
- identical augmentation;
- identical backbone;
- identical classifier;
- identical optimizer;
- identical batch size;
- identical number of epochs;
- identical learning-rate schedule;
- identical fine-tuning procedure; and
- identical evaluation metrics.

Only the experimental factor under investigation should change.

For example, when studying attention position:

$$\boxed{\text{Only Attention Position Should Change}}$$

and when studying attention type:

$$\boxed{\text{Only Attention Type Should Change.}}$$

35. Minimum Mandatory Ablation Set

If laboratory time is limited, every student must complete at least the following experiments:

1. Baseline pretrained ResNet50.
2. Channel attention at P_3.
3. Spatial attention at P_3.
4. Channel+Spatial attention at P_3.

5. Channel+Spatial attention at P_2.
6. Channel+Spatial attention at P_4.
7. Self-attention at P_3.
8. Grad-CAM comparison between baseline ResNet50 and one selected attention model.
9. Spatial-attention visualization at P_2, P_3, and P_4.
10. Graphical comparison of the classification performance of the attention models.

This minimum set allows students to study:

Attention Type

and

Attention Position.

36. Additional Student Exploration

Students interested in extended experimentation should select at least two mechanisms from:

SE, ECA, CBAM, Coordinate Attention, Self-Attention

and integrate them with the same ResNet50 backbone.

Students should maintain the same:

- dataset split;
- preprocessing;
- augmentation;
- optimizer;
- epochs;
- fine-tuning strategy; and
- attention position.

The additional investigation should include both:

Quantitative Comparison

and

Qualitative Comparison.

37. Questions Students Must Answer

1. **What is the purpose of residual connections in ResNet50?**
   Answer: Residual connections allow information and gradients to flow more easily through the network, helping it learn deeper features.
2. **What does conv2_x represent compared with conv5_x?**
   Answer: `conv2_x` learns basic features like edges and textures, while `conv5_x` learns complex, high-level features.
3. **Why does the spatial resolution decrease with network depth?**
   Answer: Spatial resolution decreases to reduce computation and help the network focus on more meaningful features.
4. **Why does the number of feature channels increase with network depth?**
   Answer: More channels allow the network to represent a wider variety of complex features.
5. **What is meant by low-level, intermediate, and high-level features?**
   Answer: Low-level features include edges and textures, intermediate features include shapes and patterns, and high-level features represent complex objects or semantic information.
6. **What is the purpose of an attention mechanism?**
   Answer: Attention helps the model focus on important features or image regions while reducing the influence of less relevant information.
7. **What does channel attention attempt to learn?**
   Answer: Channel attention learns which feature channels are most important for making predictions.
8. **What does spatial attention attempt to learn?**
   Answer: Spatial attention identifies the image locations that contain the most useful information.

9. **How is channel attention different from spatial attention?**
   Answer: Channel attention selects important feature types, whereas spatial attention selects important locations in the feature map.
10. **Why can channel and spatial attention be used sequentially?**
    Answer: Channel attention selects useful features, and spatial attention then identifies where those features are most relevant.
11. **Why might channel attention behave differently at conv3_x and conv5_x?**
    Answer: `conv3_x` contains more local patterns, while `conv5_x` contains more abstract semantic features. Therefore, the important channels may differ between stages.
12. **Why can spatial attention be useful at an intermediate ResNet stage?**
    Answer: Intermediate stages preserve useful spatial details while learning meaningful patterns, helping attention locate important image regions.
13. **Why is self-attention computationally expensive when the feature map has high spatial resolution?**
    Answer: Self-attention compares spatial positions with one another, requiring considerable computation and memory when there are many positions.
14. **What happens to the size of the self-attention matrix when the spatial resolution increases?**
    Answer: For N spatial positions, the attention matrix contains N^2 elements. Doubling the number of positions makes the matrix four times larger.
15. **How does attention performance vary when the insertion position is changed?**
    Answer: Performance depends on the insertion stage because different stages capture different features. Some positions improve accuracy more than others.
16. **How does attention at P_2 differ visually from attention at P_4?**
    Answer: P_2 generally emphasizes fine details and textures, whereas P_4 focuses more on coarse structures and higher-level semantic information.
17. **Does applying attention at multiple stages always improve classification performance?**
    Answer: No. Multiple attention modules may increase computation and overfitting without providing additional accuracy.
18. **Does Channel→Spatial attention behave differently from Spatial→Channel attention?**
    Answer: Yes. Channel→Spatial first selects important channels and then focuses on relevant locations. Spatial→Channel performs these operations in reverse order.
19. **How does the spatial-attention kernel size influence the attention map?**
    Answer: A larger kernel captures broader spatial context, while a smaller kernel focuses on more local details.
20. **How does SE attention differ from ECA attention?**
    Answer: SE uses fully connected layers to learn channel importance, while ECA uses a lightweight one-dimensional convolution to capture local channel relationships with fewer parameters.
21. **How does CBAM combine channel and spatial attention?**
    Answer: CBAM applies channel attention followed by spatial attention, refining both the important feature channels and image locations.
22. **How does self-attention differ from channel attention?**
    Answer: Self-attention models relationships between spatial positions, while channel attention determines the importance of different feature channels.
23. **Does the attention mechanism with the largest number of parameters necessarily provide the highest accuracy?**
    Answer: No. More parameters do not guarantee better accuracy. A lightweight attention mechanism may achieve similar or better results with less computation.
24. **What is the difference between an internal attention map and Grad-CAM?**
    Answer: An internal attention map shows what the attention module emphasizes, whereas Grad-CAM uses gradients to highlight image regions influencing a particular class prediction.
25. **Why is Grad-CAM called a post-hoc explanation technique?**
    Answer: Grad-CAM explains predictions after the model has been trained, without requiring the model to be specifically trained to generate explanations.
26. **Does the attention model suppress irrelevant background regions?**
    Answer: Attention can reduce the influence of irrelevant background regions, but it does not always remove them completely.
27. **Are the attention maps of correctly classified and incorrectly classified images different?**
    Answer: They can differ. Correct predictions may focus on relevant flower features, while incorrect predictions may focus on misleading regions. However, this is not always the case.
28. **Which flower classes benefit most from attention?**
    Answer: The classes showing the greatest improvement in class-wise precision, recall, or F1-score benefit most. The exact classes must be identified from the experimental results.
29. **What information can be obtained from a confusion matrix that cannot be obtained from overall accuracy alone?**

Answer: A confusion matrix reveals which classes are frequently misclassified as other classes, whereas overall accuracy only gives the total proportion of correct predictions.

30. **Why is macro F1 useful in addition to accuracy?**
    Answer: Macro F1 gives equal importance to every class, making it useful when class frequencies or classification performance vary.
31. **What is meant by the accuracy–complexity trade-off?**
    Answer: It refers to balancing classification accuracy against model complexity, parameter count, and computational requirements.
32. **What is meant by the F1–inference-time trade-off?**
    Answer: It describes the balance between classification quality and prediction speed. A model with slightly lower F1-score may still be useful if it predicts much faster.
33. **Why must every ablation study use the same dataset split?**
    Answer: Using the same dataset split ensures a fair comparison because performance differences are less likely to result from different training and testing samples.
34. **Why must the same training protocol be maintained when comparing different attention mechanisms?**
    Answer: Keeping the training protocol consistent controls factors such as epochs, optimizer, learning rate, and data augmentation, making the comparison fair.
35. **What conclusions can be drawn from the Grad-CAM comparison?**
    Answer: Grad-CAM comparisons help show whether models focus on meaningful flower regions and provide interpretable predictions. However, visual explanations alone do not prove higher accuracy or causal reasoning.

38. Observations to be Recorded

Students should record the following during experimentation:

1. **Input and output feature dimensions for each attention module:** The dimensions of feature representations ($H \times W \times C$) at the candidate insertion stages within the ResNet50 backbone were observed as follows:
   - Stage P_1 (after `conv2_x`): Input tensor shape is $56 \times 56 \times 256$, with output shape $56 \times 56 \times 256$.
   - Stage P_2 (after `conv3_x`): Input tensor shape is $28 \times 28 \times 512$, with output shape $28 \times 28 \times 512$.
   - Stage P_3 (after `conv4_x`): Input tensor shape is $14 \times 14 \times 1024$, with output shape $14 \times 14 \times 1024$.
   - Stage P_4 (after `conv5_x`): Input tensor shape is $7 \times 7 \times 2048$, with output shape $7 \times 7 \times 2048$.

   Inside the attention modules at P_3 ($14 \times 14 \times 1024$):
   - *Channel Attention (SE):* Global average pooling compresses spatial coordinates $14 \times 14 \times 1024 \to 1 \times 1 \times 1024$. The reduction FC layer with ratio $r = 16$ outputs $1 \times 1 \times 64$, followed by ReLU and an expansion FC layer outputting $1 \times 1 \times 1024$. Sigmoid activation produces the channel modulation vector $M_c \in \mathbb{R}^{1024}$, broadcast back to $14 \times 14 \times 1024$.
   - *Spatial Attention:* Along the channel axis, average-pooling and max-pooling produce two 2D feature maps concatenated into $14 \times 14 \times 2$. A single 7×7 convolution maps this to $14 \times 14 \times 1$, followed by sigmoid gating to produce spatial mask $M_s \in \mathbb{R}^{14 \times 14}$, broadcast back to $14 \times 14 \times 1024$.
   - *Self-Attention:* The feature tensor is flattened into $N = 14 \times 14 = 196$ spatial tokens. Query and Key projections map to $\mathbb{R}^{196 \times 128}$, Value maps to $\mathbb{R}^{196 \times 1024}$, computing a full spatial affinity matrix $S \in \mathbb{R}^{196 \times 196}$. The modulated projection is reshaped to $14 \times 14 \times 1024$ and added residually.
   - *ECA Attention:* Global average pooling produces $1 \times 1 \times 1024$, which is processed by a fast 1D convolution with an adaptive kernel $k = 5$ across channels, maintaining dimension $1 \times 1 \times 1024$.
   - *Coordinate Attention:* Features are pooled using 1D horizontal pooling ($14 \times 1 \times 1024$) and 1D vertical pooling ($1 \times 14 \times 1024$), concatenated and mapped via shared 1×1 convolution, generating directional coordinate attention maps that maintain $14 \times 14 \times 1024$.
2. **Training accuracy:** During Phase 1 (epochs 1–2) when the ResNet50 backbone was frozen and only the attention layers and classification head were trained, training accuracy started between 73.5% and 78.2%. Upon initiating fine-tuning in Phase 2 (epochs 3–10) with unfrozen top stages, training accuracy accelerated rapidly past 89% and converged cleanly by epoch 10. The recorded final training accuracies are:
   - Baseline ResNet50: 95.82%
   - Coordinate Attention (P_3): 95.54%
   - Self-Attention (P_3): 95.31%
   - Efficient Channel Attention (ECA, P_3): 95.22%
   - Channel Attention (SE, P_3): 95.10%
   - CBAM Attention (P_3): 94.92%
   - Spatial Attention ($k = 7, P_3$): 94.81%
   - Channel+Spatial (P_3): 94.63%
3. **Validation accuracy:** Validation accuracy climbed steadily during training, achieving stability between epochs 6 and 10 without volatile oscillations. The final converged validation accuracies observed are:

- Baseline ResNet50: 93.28%
- Spatial Attention ($k = 5, P_3$): 93.47%
- Coordinate Attention (P_3): 92.92%
- Channel+Spatial (P_2): 92.74%
- Channel+Spatial (P_4): 92.74%
- Self-Attention (P_3): 92.74%
- ECA Attention (P_3): 92.74%
- Spatial Attention ($k = 3, P_3$): 92.56%
- Channel Attention (SE, P_3): 92.38%
- Spatial Attention ($k = 7, P_3$): 92.38%
- CBAM Attention (P_3): 92.38%
- Channel+Spatial (P_3): 92.20%

4. **Testing accuracy:** Testing accuracy was recorded on the unseen test split consisting of 551 images from the `tf_flowers` dataset:
   - Baseline ResNet50: **93.28%** (514 / 551 correctly classified)
   - Spatial Attention ($k = 5, P_3$): **93.47%** (515 / 551)
   - Coordinate Attention (P_3): **92.92%** (512 / 551)
   - Channel+Spatial at P_2: **92.74%** (511 / 551)
   - Channel+Spatial at P_4: **92.74%** (511 / 551)
   - Self-Attention at P_3: **92.74%** (511 / 551)
   - ECA Attention at P_3: **92.74%** (511 / 551)
   - Multi-stage Attention ($P_2 + P_3 + P_4$): **92.92%** (512 / 551)
   - Spatial Attention ($k = 3, P_3$): **92.56%** (510 / 551)
   - Spatial $\rightarrow$ Channel (P_3): **92.38%** (509 / 551)
   - Channel Attention (SE, P_3): **92.38%** (509 / 551)
   - Spatial Attention ($k = 7, P_3$): **92.38%** (509 / 551)
   - CBAM Attention at P_3: **92.38%** (509 / 551)
   - Multi-stage Attention ($P_3 + P_4$): **92.20%** (508 / 551)
   - Channel+Spatial at P_3: **92.20%** (508 / 551)
5. **Training loss:** Cross-entropy training loss ($\mathcal{L}_{CE}$) initiated at 0.78–0.86 during epoch 1 and experienced a rapid monotonic descent once fine-tuning unlocked deeper convolutional weights at epoch 3. Terminal training loss settled between 0.14 and 0.17 across all models, with Baseline reaching 0.142, Coordinate Attention reaching 0.147, Self-Attention reaching 0.151, and Channel+Spatial reaching 0.168. Training dynamics were remarkably stable across all runs with no numerical instabilities or exploding gradients.
6. **Validation loss:** Validation loss descended from $\approx$ 0.68 in epoch 1 to a stable plateau between 0.28 and 0.34 across epochs 6–10. Baseline ResNet50 converged to the lowest validation loss of 0.285. Among attention models, Coordinate Attention (0.298) and ECA (0.304) maintained the lowest and smoothest validation losses. Channel+Spatial (P_3) leveled off at 0.321. The training-to-validation loss gap remained narrow ($\Delta \approx 0.14$–0.16) across all configurations, confirming that the attention modules regularized learning effectively without severe overparameterization.
7. **Macro precision:** Macro precision reflects the unweighted average of precision across all five classes, preventing larger classes from dominating the metric:
   - Baseline ResNet50: **93.61%**
   - Coordinate Attention (P_3): **93.42%**
   - Self-Attention (P_3): **93.06%**
   - Channel+Spatial (P_2): **93.04%**
   - Channel+Spatial (P_4): **93.02%**
   - ECA Attention (P_3): **92.97%**
   - Channel Attention (SE, P_3): **92.80%**
   - CBAM Attention (P_3): **92.67%**
   - Spatial Attention ($k = 7, P_3$): **92.58%**
   - Channel+Spatial (P_3): **92.50%**
8. **Macro recall:** Macro recall measures the average sensitivity across classes:
   - Baseline ResNet50: **93.34%**
   - Coordinate Attention (P_3): **92.77%**
   - Channel+Spatial (P_4): **92.72%**
   - Channel+Spatial (P_2): **92.66%**
   - Self-Attention (P_3): **92.65%**
   - ECA Attention (P_3): **92.63%**
   - Spatial Attention ($k = 7, P_3$): **92.49%**
   - CBAM Attention (P_3): **92.42%**
   - Channel Attention (SE, P_3): **92.35%**

- Channel+Spatial (P_3): **92.11%**

9. **Macro F1-score:** Macro F1 harmonic mean between precision and recall was recorded as:
   - Spatial Attention ($k = 5, P_3$): **93.53%**
   - Baseline ResNet50: **93.37%**
   - Coordinate Attention (P_3): **92.98%**
   - Channel+Spatial (P_4): **92.78%**
   - Channel+Spatial (P_2): **92.76%**
   - Self-Attention (P_3): **92.76%**
   - ECA Attention (P_3): **92.73%**
   - Spatial Attention ($k = 3, P_3$): **92.62%**
   - Channel Attention (SE, P_3): **92.47%**
   - Spatial Attention ($k = 7, P_3$): **92.42%**
   - CBAM Attention (P_3): **92.34%**
   - Channel+Spatial (P_3): **92.22%**
10. **Number of trainable parameters:**
    - *Initial frozen phase (Stage 1):* Only the attention module and classification head were trainable:
      - Classification Dense Head: $1024 \times 5 + 5 = 5,125$ parameters.
      - Squeeze-and-Excitation at P_3: $1024 \times 64 + 64 + 64 \times 1024 + 1024 = 132,160$ parameters (≈ 0.13 M).
      - Spatial Attention ($k = 7$): $2 \times 7 \times 7 \times 1 + 1 = 99$ parameters.
      - ECA Attention: 1D Conv with kernel $k = 5$ across 1 channel $= 5 \times 1 + 1 = 6$ parameters.
      - Self-Attention at P_3: Linear projections θ, ϕ, g and output projection $= 1,313,281$ parameters (≈ 1.31 M).
      - Coordinate Attention at P_3: Shared 1×1 convolution and batch normalization $= 98,304$ parameters (≈ 0.10 M).
    - *Fine-tuning phase (Stage 2):* Unfreezing the final stage of ResNet50 (`conv5_x`) added $14,961,408$ trainable parameters across all architectures.
11. **Total number of model parameters:** The complete parameter footprint for each evaluated architecture was recorded as:
    - Baseline ResNet50: **23.52 M** ($23,518,085$ parameters)
    - Spatial Attention ($k = 7, P_3$): **23.52 M** (+99 parameters)
    - ECA Attention (P_3): **23.52 M** (+6 parameters)
    - Channel+Spatial at P_2: **23.55 M** (at $C = 512$, adds ≈ 0.03 M)
    - Coordinate Attention (P_3): **23.62 M** (+0.10 M)
    - Channel Attention (SE, P_3): **23.65 M** (+0.13 M)
    - CBAM Attention (P_3): **23.65 M** (+0.13 M)
    - Channel+Spatial at P_4: **24.04 M** (at $C = 2048$, adds ≈ 0.52 M)
    - Self-Attention (P_3): **24.83 M** (+1.31 M)
12. **Average inference time:** Single-image inference latency was benchmarked with batch size 1:
    - Channel Attention (SE, P_3): **19.76 ms/img** (fastest execution due to highly vectorizable 1D vector scaling)
    - ECA Attention (P_3): **22.06 ms/img**
    - Channel+Spatial (P_4): **22.08 ms/img** (small 7×7 spatial resolution yields minimal convolution overhead)
    - Baseline ResNet50: **22.37 ms/img**
    - Spatial Attention ($k = 7, P_3$): **22.41 ms/img**
    - Self-Attention (P_3): **22.80 ms/img**
    - Coordinate Attention (P_3): **23.28 ms/img**
    - Channel+Spatial (P_2): **23.32 ms/img**
    - Channel+Spatial (P_3): **23.44 ms/img**
    - CBAM Attention (P_3): **23.99 ms/img** (slowest due to consecutive channel pooling, MLP, spatial pooling, and 7×7 convolution)
13. **Confusion matrix:** The confusion matrices evaluated on the 551 test samples revealed:
    - *Baseline ResNet50 (*93.28%*):* Daisy: 89 correct, 6 misclassified as dandelion; Dandelion: 130 correct, 1 as daisy, 3 as sunflowers, 1 as tulips; Roses: 91 correct, 2 as dandelion, 1 as sunflowers, 2 as tulips; Sunflowers: 100 correct, 1 as daisy, 3 as dandelion, 1 as roses; Tulips: 104 correct, 1 as daisy, 4 as dandelion, 10 as roses, 1 as sunflowers.
    - *Channel → Spatial (P_3) (*92.20%*):* Daisy: 88 correct, 6 as dandelion, 1 as tulips; Dandelion: 131 correct (+1), 1 as daisy, 3 as sunflowers; Roses: 87 correct (-4), 2 as dandelion, 7 as tulips (+5); Sunflowers: 100 correct, 4 as dandelion, 1 as roses; Tulips: 102 correct (-2), 1 as daisy, 3 as dandelion, 13 as roses (+3), 1 as sunflowers.
    - *Dominant Error Pattern:* Mutual confusion between Roses and Tulips is the primary error source in both networks (10 to 13 tulips misclassified as roses, and 2 to 7 roses misclassified as tulips) due to overlapping crimson/scarlet petal coloration, organic petal curvatures, and green leafy backdrops. The attention model at P_3 incurred +5 additional misclassifications of roses as tulips, which accounts entirely for the minor overall accuracy drop of 1.08%.
14. **Channel-attention weights:**

- Learned channel weights $M_c \in \mathbb{R}^{1024}$ at P_3 span continuously from 0.52 to 0.98, forming a unimodal distribution centered predominantly in $[0.75, 0.95]$.
- A consistent subset of approximately 20 channels receives top weights > 0.92 across all test images (top indices include 950, 803, 826, 245, 464, 19, 790, 206, 979, and 966), indicating that the network relies on specialized feature channels.
- Dynamic input-dependent modulation was observed: dandelion inputs heavily weight high-frequency texture and yellow-chroma feature channels, whereas tulip and rose inputs prioritize smooth contour and red-spectrum feature channels.

15. **Spatial-attention maps:**
    - Sigmoidal spatial attention values M_s at P_3 fall in $[0.89, 0.97]$.
    - Across all five flower categories, spatial attention reliably peaks (0.94–0.97) over the central floral disc, petals, and reproductive anatomy, while assigning lower attention (0.89–0.91) to background grass, soil, and stems.
    - Receptive field comparison confirmed that kernel size $k = 5$ provides optimal context-aware boundary localization $[0.85, 0.95]$, whereas $k = 3$ is overly fragmented $[0.80, 0.92]$ and $k = 7$ is overly smooth $[0.90, 0.96]$.
16. **Self-attention responses:**
    - The 196×196 spatial affinity matrix reveals pronounced non-local clustering across distant coordinates.
    - Querying a single petal coordinate (e.g., query at $(7, 7)$ or $(3, 3)$) produces high affinity values across all other radially symmetric petals of the flower, while assigning near-zero affinity to surrounding leaves and soil.
    - Aggregated mean and max query responses cleanly envelope the complete silhouette of the flower, confirming global contextual awareness.
17. **Grad-CAM maps:**
    - Baseline ResNet50 produces relatively diffuse activation heatmaps that frequently spread into background foliage, stems, and soil.
    - Attention-equipped architectures (Channel, Spatial, CBAM, Coordinate Attention) generate sharply focused, high-contrast heatmaps tightly concentrated on discriminative petal arrangements and floral centers.
    - On misclassified test samples (e.g., daisy misclassified as dandelion), Grad-CAM revealed that the network over-attended to the yellow central disc and failed to attend to the distinguishing white ray petals.
18. **Attention maps for different insertion positions (P_1, P_2, P_3, P_4):**
    - At P_1 (56×56, after `conv2_x`): Attention responds to high-frequency edge gradients, fine petal margins, and sharp color contrasts ($M_s \in [0.88, 0.95]$).
    - At P_2 (28×28, after `conv3_x`): Attention groups intermediate structural parts, outlining individual petal clusters and calyx structures ($M_s \in [0.89, 0.96]$).
    - At P_3 (14×14, after `conv4_x`): Attention achieves balanced part-to-object abstraction, cleanly separating the main floral blossom from surrounding foliage ($M_s \in [0.89, 0.98]$).
    - At P_4 (7×7, after `conv5_x`): Attention condenses into a single coarse semantic blob covering the general center of mass of the flower ($M_s \in [0.91, 1.00]$), losing boundary delineation due to aggressive spatial downsampling.

39. Suggested Laboratory Report Structure

The laboratory report may be organized as follows:

1. **Aim**
2. **Dataset Description**
3. **ResNet50 Architecture**
4. **Attention Mechanisms**
5. **Proposed Experimental Architecture**
6. **Experimental Setup**
7. **Attention-Type Ablation**
8. **Attention-Position Ablation**
9. **Performance Results**
10. **Complexity Analysis**
11. **Training and Validation Graphs**
12. **Confusion Matrix Analysis**
13. **Grad-CAM Analysis**
14. **Internal Attention-Map Analysis**
15. **Qualitative Comparison**
16. **Additional Attention-Mechanism Comparison**
17. **Discussion**
18. **Conclusion**

40. Discussion Guidelines

Students should not merely reproduce numerical results.

The discussion should explain:

- why a particular attention mechanism produced a change in performance;
- how attention location influences feature refinement;
- how feature resolution changes attention behavior;
- how model complexity changes with different mechanisms;
- whether performance improvements justify additional computational cost;
- whether attention maps correspond to meaningful object regions;
- whether attention helps suppress the background;
- whether attention behaves differently for incorrect predictions; and
- whether quantitative and qualitative observations support each other.

Students should avoid making conclusions based on a single image.

Qualitative observations should preferably be based on several representative test images.

41. Expected Outcome

At the end of the experiment, students should understand that an attention mechanism is not an isolated module whose behavior is independent of the backbone.

Its effectiveness depends on:

$$\boxed{\begin{aligned}\text{Attention Effectiveness} = {} & \text{Attention Type} \\ & + \text{Attention Position} \\ & + \text{Feature Resolution} \\ & + \text{Feature Semantics} \\ & + \text{Model Complexity.}\end{aligned}}$$

Students should also understand that classification performance alone does not provide a complete explanation of an attention mechanism.

A comprehensive analysis should therefore combine:

$$\boxed{\text{Classification Performance}}$$

with

$$\boxed{\text{Computational Complexity}}$$

and

$$\boxed{\text{Qualitative Visualization.}}$$

Therefore:

$$\boxed{\text{Complete Attention Analysis} = \text{Quantitative} + \text{Computational} + \text{Qualitative}}$$

The final conclusions must be based on the student's experimental results rather than on assumptions about which attention mechanism should perform best.

42. Source Code

The complete source code, datasets, generated plots, and report files for this experiment are available in the following GitHub repository:

GitHub Repository:
https://github.com/VIKASNI-S/Deep-Learning-Lab/tree/main/Lab-8

43. References

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